

CMPUT 654: Mathematical Foundations of Modern AI Systems

Fall 2026

Lecture 9: Interpolation and benign overfitting

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Covered in class. The lecture opened with a brief discussion of the random-label experiments of Zhang et al., which motivated the interpolation question. The main development followed roughly [Sections 9.1 to 9.3](#), up to the statement of [Theorem 9.3.1](#); there was no time to prove the theorem. The result for smooth functions in [Corollary 9.3.3](#) was also mentioned, and we discussed the problem with expected risk and the use of clipping. The theorem’s proof, the kernel results, and the extended discussion of generalization theory are supplementary material in these notes.

Reading guide. Motivated by exact fitting in large neural networks, these notes ask when interpolation of noisy observations can coexist with consistent prediction. We first determine how large the benign set can be by a point-correction argument. We then use minimum-norm linear interpolation as a tractable proxy for gradient training and analyze one polynomial procedure: Legendre polynomials represent the low-degree signal, and bounded Chebyshev polynomials provide the additional interpolation directions. Both are functions of the same one-dimensional input. The main text derives the approximation and estimation bound; [Appendix 9.B](#) verifies its matrix assumptions. [Sections 9.4](#) and [9.5](#) extend the discussion: they compare positive and negative results for kernel interpolation and ask what they justify about model size and training time. [Section 9.4](#) first introduces kernels and reproducing kernel Hilbert spaces through linear function approximation. The optional discussion in [Section 9.6](#) examines what the random-label experiments establish about generalization theory. [Appendix 9.C](#) explains why the polynomial guarantee in probability does not control expected risk and how clipping repairs this failure. The prerequisites are least squares, orthogonal projections, elementary matrix algebra, and convergence in probability. [Appendix 9.A](#) reviews the function-space terminology, and the exercises ask you to supply the linear algebra used in the main proof.

9.1 The question raised by interpolation

Large neural networks trained by gradient methods commonly reach zero training error. In the experiments of [Zhang et al. \[2017\]](#), the same networks could also fit arbitrarily randomized labels, while training on the original labels gave useful test predictions. Thus the model and optimizer had enough capacity to fit pure noise, yet exact fitting on the natural problem remained compatible with prediction. This observation motivates the advice to use large models and train them to interpolation. We ask when that advice describes a sound statistical procedure.

We begin with a one-dimensional regression analogue. Suppose that

$$Y_i = f^*(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid X_i] = 0.$$

When the training inputs are distinct, an interpolating rule satisfies $\hat{f}_n(X_i) = Y_i$ at every one of them. It therefore reproduces each realized noise value there. Can the same rule still estimate f^* accurately at a new input?

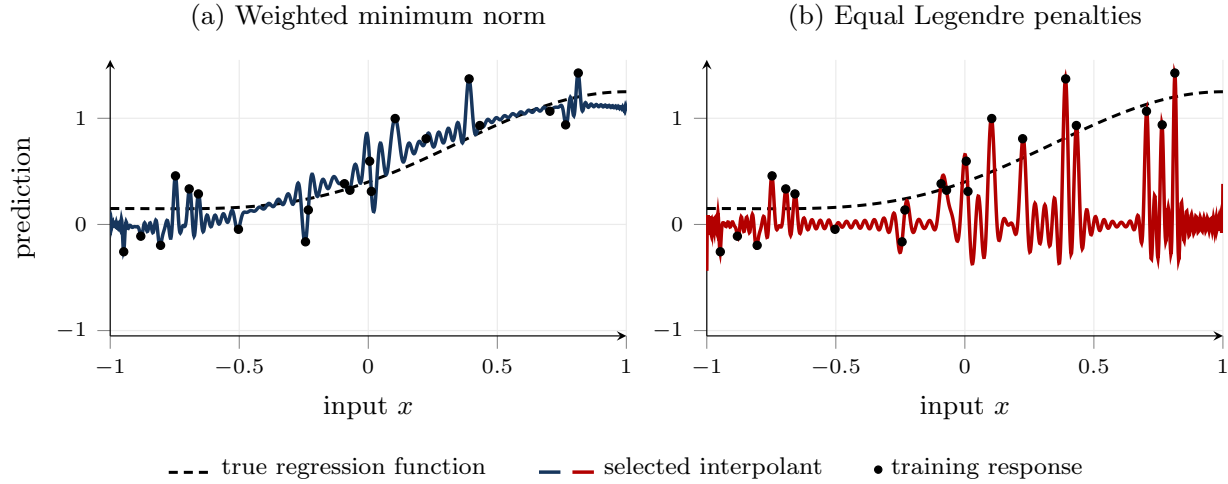


Figure 1: Both panels use the same $n = 20$ observations from $X \sim \text{Unif}[-1, 1]$ and $Y = f^*(X) + \mathcal{N}(0, 0.28^2)$, where $f^*(x) = 0.55 + 0.55 \sin(\pi x/2) - 0.15 \cos(\pi x)$, and the same space of dimension 164. Both rules interpolate. The weighted norm distinguishes the first $m = 4$ features; equal Legendre penalties minimize $L^2(\mu)$ norm.

Figure 1 shows both outcomes on one sample. The weighted minimum-norm rule follows the smooth nonpolynomial regression function between the observations. Equal penalties on the Legendre coefficients select a function that stays close to zero over most of the domain and makes narrow excursions to the observed responses. Both rules use the same polynomial space and interpolate the same data. The selection rule changes the population behavior. This comparison directs attention from the size of the available model to the procedure that selects one interpolant. Later sections establish asymptotic versions of both outcomes.^{1 2}

We therefore study the following descriptive question:

For which fixed data-generating laws does a specified interpolating learning procedure have vanishing population excess risk?

To make the question precise, fix a nonempty measurable input space $\mathcal{X}$, and let $\mathcal{P}_2$ be the class of probability laws P on $\mathcal{X} \times \mathbb{R}$ satisfying $\mathbb{E}_P[Y^2] < \infty$. Let $S_n = ((X_1, Y_1), \dots, (X_n, Y_n))$ be an i.i.d. sample from a fixed $P \in \mathcal{P}_2$. Write $\mu = P_X$ and

$$f_P^*(x) = \mathbb{E}[Y \mid X = x].$$

For functions, $\|f\|_2$ abbreviates $\|f\|_{L^2(\mu)}$ for the current input law μ . For an independent test pair (X, Y) with law P and a fixed predictor $f \in L^2(\mu)$, the squared-loss identity

$$\mathbb{E}[(f(X) - Y)^2] - \mathbb{E}[(f_P^*(X) - Y)^2] = \|f - f_P^*\|_{L^2(\mu)}^2 \quad (9.1)$$

shows that learning the conditional mean is exactly the problem of making the $L^2(\mu)$ error small. For a predictor selected from S_n , apply this identity conditional on S_n . Its excess risk is then a random variable determined by the training sample. [Exercise 1](#) asks you to prove both claims.

Definition 9.1.1 (Interpolating procedure and the benign set). Let $B = (B_n)_{n \geq 1}$ be a fixed learning procedure, including any predetermined choices of model dimension, feature scaling, norm, and

optimization rule. In a sample S_n , call index i *nonconflicting* if $X_j = X_i$ implies $Y_j = Y_i$ for every j . For fixed n , call B_n *interpolating* if, for every sample,

$$B_n(S_n)(X_i) = Y_i, \quad \text{for every nonconflicting index } i.$$

Call B interpolating if every B_n is interpolating. For such a procedure, define

$$\mathcal{A}(B) = \left\{ P \in \mathcal{P}_2 : \|B_n(S_n) - f_P^*\|_{L^2(P_X)}^2 \xrightarrow{\mathbb{P}} 0 \text{ as } n \rightarrow \infty \right\}. \quad (9.2)$$

For $P \in \mathcal{A}(B)$, we say that the overfitting performed by B on P is *benign*.

Repeated copies with the same response are nonconflicting; only a group that assigns different responses to the same input is omitted. The definition allows arbitrary input laws. Each optimization procedure below is understood to impose constraints only at nonconflicting inputs.

On noisy problems, *overfitting* refers here to exact fitting of realized response noise. The word *benign* refers to the vanishing population excess risk. Thus interpolation names the fitting requirement, while *benign overfitting* names the combination. This is the standard term in the recent literature.³

How strong is it to prove that $\mathcal{A}(B)$ is nonempty? By itself, this is a vacuous requirement.

Proposition 9.1.2 (Why nonemptiness alone is automatic). *Every interpolating procedure B satisfies $\mathcal{A}(B) \neq \emptyset$.*

Proof. Fix x_0 and y_0 , and let P assign probability one to (x_0, y_0) . Every sample consists only of copies of this pair. Interpolation forces $B_n(S_n)(x_0) = y_0$, while $f_P^*(x_0) = y_0$. Since the test input equals x_0 almost surely, the population excess risk is zero for every n . $\square$

The interpolation question is most informative when the responses contain nondegenerate noise, meaning

$$\sigma_P^2 := \mathbb{E}[(Y - f_P^*(X))^2] > 0.$$

If $Y = f_P^*(X)$ almost surely, every observed response is a target value, so exact fitting creates no direct conflict with estimating the target. A poor selection rule can still fail between observations. With response noise, interpolation additionally forces the procedure to fit random deviations from the target. We therefore seek noisy members and nonmembers of benign sets.

9.2 Interpolation is compatible with L^2 approximation

Can interpolation preserve the accuracy of an estimator that already generalizes? The setup asks for an interpolating procedure and vanishing population error, but these properties concern different inputs: exact fitting concerns the observations, while integrated loss concerns a fresh input. Start with a consistent kernel smoother, or any other consistent regression estimator, and modify its values only at the observations.

Proposition 9.2.1 (Point-corrected smoother). *Suppose that $\mu(\{x\}) = 0$ for every $x \in \mathcal{X}$, and that $\tilde{f}_n$ is any estimator satisfying*

$$\|\tilde{f}_n - f_P^*\|_{L^2(\mu)} \xrightarrow{\mathbb{P}} 0 \quad \text{as } n \rightarrow \infty.$$

For an arbitrary sample, let $I_n(S_n)$ contain one representative of every distinct nonconflicting input, and define

$$\hat{f}_n(x) = \tilde{f}_n(x) + \sum_{i \in I_n(S_n)} (Y_i - \tilde{f}_n(X_i)) \mathbb{I}\{x = X_i\}. \quad (9.3)$$

Then the resulting procedure is interpolating, and

$$\|\hat{f}_n - f_P^*\|_{L^2(\mu)} = \|\tilde{f}_n - f_P^*\|_{L^2(\mu)} \xrightarrow{\mathbb{P}} 0 \quad \text{as } n \rightarrow \infty.$$

Proof. For every sample, substituting a nonconflicting input into Eq. (9.3) gives $\hat{f}_n(X_i) = Y_i$. Thus the procedure is interpolating. The assumption on μ makes every finite set have measure zero. The difference $\hat{f}_n - \tilde{f}_n$ is supported on the finite set $\{X_1, \dots, X_n\}$. Therefore its $L^2(\mu)$ norm is zero. In particular, the triangle inequality gives

$$\begin{aligned} \|\hat{f}_n - f_P^*\|_{L^2(\mu)} &\leq \|\tilde{f}_n - f_P^*\|_{L^2(\mu)} + \|\hat{f}_n - \tilde{f}_n\|_{L^2(\mu)} \\ &= \|\tilde{f}_n - f_P^*\|_{L^2(\mu)}. \end{aligned}$$

The reverse inequality holds because the two functions agree μ -almost everywhere. $\square$

The added terms change only the observed point values.⁴ The result proves that interpolation and L^2 consistency can coexist. Its selection rule is an explicit point correction; gradient training of a neural network uses a different rule. In Exercise 2, you are asked to show that averaging the responses observed at each distinct input extends the construction to arbitrary input laws under the interpolation convention in Definition 9.1.1.

Can the benign set equal the full problem class? Let $\mathcal{C} \subseteq \mathcal{P}_2$ be a class on which one fixed base procedure $\tilde{B}$ is L^2 -consistent. Point correction produces one interpolating procedure B with $\mathcal{C} \subseteq \mathcal{A}(B)$ when the laws in $\mathcal{C}$ have no point masses; response averaging gives the same conclusion for arbitrary input laws. In particular, a universally consistent base procedure yields an interpolating procedure whose benign set is the full problem class on which universality holds. Standard universally consistent regression procedures, with truncation for unbounded responses, provide such bases on Euclidean input spaces.

Consequently, no negative theorem can show that every interpolating learning procedure fails on some law in such a class. A negative result must restrict the learning procedure, its output geometry, or the required mode of risk control.

9.3 A minimum-norm polynomial interpolator

Can a neural network trained by gradient descent to interpolation have a nontrivial benign set $\mathcal{A}(B)$? The predictor selected by gradient descent in a finite nonlinear network is difficult to characterize. We therefore study the corresponding linear problem, where the selected interpolant can be described exactly.

In the linear setting, the link with gradient training is exact. In Problem 3 of Homework 0, you were asked to prove that, for a full-row-rank design matrix, gradient descent on the squared loss, started at zero with a sufficiently small constant step size, converges to the interpolating parameter vector of smallest Euclidean norm. Gradient flow started at zero selects the same vector. The next lecture will develop this connection further. Minimum-norm interpolation is therefore the natural linear analogue of interpolation by gradient training. Formally, let $\varphi : \mathcal{X} \rightarrow \mathbb{R}^p$ be some fixed function, collecting the $p \geq n$ basis functions to be used. A linear combination of the underlying basis functions takes the form

$$f_\theta(x) = \theta^\top \varphi(x) \quad (x \in \mathcal{X}).$$

The minimum norm interpolant is then $f_{\hat{\theta}}$ where $\hat{\theta}$ solves the problem

$$\min \|\theta\| \quad \text{subject to} \quad \theta \in \mathbb{R}^p, \quad f_\theta(X_i) = Y_i, \quad 1 \leq i \leq n. \quad (9.4)$$

Above, $\|\cdot\|$ is the 2-norm, because this is what is selected for by gradient descent. The problem has a solution as long as the $n \times p$ matrix whose (i, j) th element is $\varphi_j(X_i)$ has rank n . For now, assume this is the case. The solution is unique whenever at least one solution exists. [Exercise 3](#) asks you to prove uniqueness and derive the minimum-norm solution.

While the norm above is fixed to the 2-norm, by linearly combining the basis functions, one can change what interpolant will be selected, while leaving the set of functions that can be represented intact. To see how this works let G be an invertible $p \times p$ matrix and consider replacing φ above with $\varphi^{(G)}(x) := G\varphi(x)$. Correspondingly, let $f_c^{(G)}(x) := c^\top \varphi^{(G)}(x)$. On the one hand,

$$\mathcal{F} := \{f_\theta : \theta \in \mathbb{R}^p\} = \{f_c^{(G)} : c \in \mathbb{R}^p\},$$

while for all $\theta \in \mathbb{R}^p$,

$$f_\theta(x) = \theta^\top G^{-1}G\varphi(x) = (G^{-T}\theta)^\top \varphi^{(G)}(x) = f_{G^{-T}\theta}^{(G)}(x).$$

Hence, the norm $\|\theta\|$ becomes $\|G^{-T}\theta\|$ when changing the basis functions from φ to $\varphi^{(G)}$.

One can make this more explicit by introducing a norm on $\mathcal{F}$ based on choosing a set of basis functions that span $\mathcal{F}$. In particular, for any $\psi : \mathcal{X} \rightarrow \mathbb{R}^p$ such that $\mathcal{F} = \{x \mapsto \theta^\top \psi(x) : \theta \in \mathbb{R}^p\}$, for $f \in \mathcal{F}$, let

$$\|f\|_\psi := \inf\{\|\theta\| : f(\cdot) = \theta^\top \psi(\cdot), \theta \in \mathbb{R}^p\}.$$

One can then show that $\|\cdot\|_\psi$ is indeed a norm on $\mathcal{F}$ (when $\mathcal{F}$ is viewed as a vector space over the reals with the usual operations). Also, the optimization problem in [Eq. \(9.4\)](#) is equivalent to

$$\min \|f\|_\varphi \quad \text{subject to} \quad f \in \mathcal{F}, \quad f(X_i) = Y_i, \quad 1 \leq i \leq n \quad (9.5)$$

in the sense that if f^* is a solution to this problem and θ^* is a solution to [Eq. \(9.4\)](#) then $f^* = f_{\theta^*}$. Finally, with this notation it is clear that switching from φ to $\varphi^{(G)}$ changes the optimization problem so that the norm $\|f\|_\varphi$ in [Eq. \(9.5\)](#) is replaced by $\|f\|_{\varphi^{(G)}}$. [Exercise 4](#) asks you to verify these claims, including the case of linearly dependent features.

9.3.1 Benign overfitting with polynomials

In the rest of this section we consider the case when an unknown regression function is to be approximated by polynomials with the goal of demonstrating how one can achieve benign overfitting for a reasonably large problem class. In particular, we will consider polynomial approximation of smooth functions. For simplicity, we consider the case when $\mathcal{X} = [-1, 1]$ and when $\mu = P_X$ is uniform on this domain. Retain the i.i.d. sampling setup above, write $f^* = f_p^* \in L^2(\mu)$, and assume that, for some $\sigma^2 < \infty$,

$$Y_i = f^*(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X_i] = 0, \quad \mathbb{E}[\varepsilon_i^2 | X_i] \leq \sigma^2. \quad (9.6)$$

The conditional variance may depend on the input.

For $g : \mathbb{R} \rightarrow \mathbb{R}$, recall that g is a polynomial of degree r ($\deg(g) = r$) if for every $x \in \mathbb{R}$, $g(x) = \sum_{i=0}^r a_i x^i$ with some coefficients $(a_i)_i$ with $a_r \neq 0$. Set $\deg(0) = -\infty$ so that the zero polynomial is included below. Let $\mathcal{F}_r$ denote the set of polynomials of degree below r : $\mathcal{F}_r = \{g : g \text{ is a polynomial and } \deg(g) < r\}$. Note that this is an r -dimensional subspace of $\mathbb{R}^\mathbb{R}$ (and by restricting the functions to $\mathcal{X}$ of $\mathbb{R}^\mathcal{X}$). Given $f^* \in L^2(\mu)$, define the best squared approximation error of approximating f^* with polynomials of degree below m to be

$$A_m(f^*) = \inf_{g \in \mathcal{F}_m} \|g - f^*\|_{L^2(\mu)}^2. \quad (9.7)$$

For $s > 0$, say that f^* has *polynomial approximation order s* if, for some fixed $L < \infty$,

$$A_m(f^*) \leq L^2 m^{-2s}, \quad m \geq 1. \quad (9.8)$$

Jackson approximation estimates imply this condition for the usual s -smooth classes. For example, for integer $s \geq 1$, membership in the Sobolev space $W^{s,2}([-1, 1])$ suffices: f^* and its weak derivatives through order s are square integrable. Lipschitz functions satisfy the condition with $s = 1$. Classical regression bounds balance squared approximation error of order m^{-2s} with estimation error of order m/n , giving the squared $L^2(\mu)$ error rate $n^{-2s/(2s+1)}$. Achieving this balance by polynomial least squares also requires control of the sampled design matrix. Under uniform sampling, the sufficient condition used below is $m^2 \log m = o(n)$; the balancing choice $m \asymp n^{1/(2s+1)}$ satisfies it when $s > 1/2$. The exponent is minimax optimal over bounded s -smooth classes under standard nondegenerate noise assumptions. No estimator achieves a faster rate uniformly over such a class. The bibliographical notes give references for approximation, least-squares stability, and minimax optimality separately.

The challenge we consider is whether the same rate can be obtained by using minimum-norm interpolants – even when the responses are noisy. This is a challenge because normal least-squares fitting does not interpolate. Indeed, the best polynomial (as defined in [Eq. \(9.7\)](#)) approximator, given any degree r , only depends on the f^* , hence, this polynomial will *not* interpolate. Also, standard least-squares fitting, would not interpolate. In fact it is considered a strength of least-squares fitting that it “cleans” (or “smoothens”), separating signal (f^*) from noise.

It also follows that to meet the challenge we need to make a deliberate choice of what coordinate (basis) functions we use to parametrize the chosen subspace $\mathcal{F}_p$. First, as noted beforehand, $p \geq n$ is necessary for allowing for interpolation. Our strategy will be to split $p = m + D$ into $m \ll p$ basis functions that will span $\mathcal{F}_m$ – the goal of these is to capture the signal in the data –, and a remaining set of extra D basis functions, whose purpose will be to fit to the noise. In particular, we choose the basis and scaling of these two blocks so that the selected interpolant represents the low-degree component in the first block and uses the second block for sample-specific residuals with small population L^2 norm, essentially following the strategy used by the point-corrected smoother of [Proposition 9.2.1](#).

For the first block, let P_j be the degree- j Legendre polynomial normalized by $P_j(1) = 1$, and set

$$\psi_j(x) = \sqrt{2j+1} P_j(x), \quad j \geq 0.$$

The functions $(\psi_j)_{j \geq 0}$ form an orthonormal basis of $L^2(\mu)$. [Appendix 9.A](#) reviews L^2 spaces, inner products, Hilbert spaces, and orthonormal bases. The significance of this choice is that for any $\beta = (\beta_j)_j$, $m \geq 1$, $\|\sum_{j=0}^{m-1} \beta_j \psi_j\|_{L^2(\mu)}^2 = \sum_{j=0}^{m-1} \beta_j^2 = \|\beta\|^2$. Minimizing the coefficient 2-norm is the same as minimizing the $L^2(\mu)$ norm of the underlying function.

For the second block, define

$$\chi_0(x) = 1, \quad \chi_j(x) = \sqrt{2} T_j(x) \quad (j \geq 1), \quad T_j(\cos \theta) = \cos(j\theta).$$

The functions χ_j are uniformly bounded by $\sqrt{2}$ and orthonormal under the arcsine law $\nu(dx) = dx/(\pi\sqrt{1-x^2})$. Since $\mu \leq (\pi/2)\nu$ (in the sense that $\mu(A) \leq (\pi/2)\nu(A)$ for any $A \subset \mathbb{R}$ Borel),

$$\left\| \sum_j b_j \chi_j \right\|_{L^2(\mu)}^2 \leq \frac{\pi}{2} \sum_j b_j^2 \quad (9.9)$$

for every finite coefficient vector. [Exercise 5](#) asks you to justify both this inequality and the orthonormal norm identity. Thus the first block has exact population-norm accounting, while the second has the upper bound needed to control the population cost of interpolation.

Combining the two blocks gives the feature map

$$\phi_{m,D}(x) = (\psi_0(x), \dots, \psi_{m-1}(x), D^{-1/2}\chi_m(x), \dots, D^{-1/2}\chi_{m+D-1}(x))^\top. \quad (9.10)$$

The factor $D^{-1/2}$ is the central design choice. The D additional features can collectively fit an arbitrary residual vector. On the sample event established below, the unscaled features can fit a residual z with coefficient squared norm at most $C\|z\|^2/D$. Without rescaling, increasing D would therefore make it progressively cheaper in parameter norm to fit the entire response with the additional block. Multiplication by $D^{-1/2}$ multiplies this parameter cost by D . A fixed low-degree component then has bounded cost in the first block and reduces the residual cost across all n observations; the sample-specific remainder is fitted by the second block. Its population squared norm remains at most $C\|z\|^2/D$.

Write a predictor in these coordinates as

$$f_{\beta,b}(x) = \sum_{j < m} \beta_j \psi_j(x) + \sum_{j=m}^{m+D-1} b_j \chi_j(x), \quad \theta = (\beta, \sqrt{D}b).$$

Then $f_{\beta,b}(x) = \theta^\top \phi_{m,D}(x)$ and

$$\|\theta\|^2 = \|\beta\|^2 + D\|b\|^2. \quad (9.11)$$

The factor D in this expression is therefore induced by the feature scaling. The empirical-loss objective remains unregularized, and the procedure minimizes this parameter norm subject to exact interpolation.

The combined features have successive polynomial degrees and nonzero leading coefficients, so they span all polynomials of degree less than $m + D$. At n distinct inputs their evaluation matrix has row rank n , provided $m + D \geq n$. The procedure therefore interpolates arbitrary responses almost surely. The remaining question is whether its population error vanishes.

9.3.2 Approximation, estimation, and interpolation cost

How accurately does this optimization procedure estimate f^* ? We use the low-degree component to estimate a polynomial approximation and show that the extra component has small population norm:

Theorem 9.3.1 (Benign overfitting for weighted polynomial interpolation). *Under the uniform-input model [Eq. \(9.6\)](#), there are universal constants $c, C > 0$ such that, if $m + D \geq n$ and $D \geq 16m$, then for every $\delta \in (0, 1)$, with probability at least*

$$1 - Cn^2/D - m \exp(-cn/m^2) - \delta,$$

the weighted minimum-norm interpolator $\hat{f}$ satisfies

$$\|\hat{f} - f^*\|_{L^2(\mu)}^2 \leq \frac{C}{\delta} \left(A_m(f^*) + \frac{\sigma^2 m}{n} + \frac{\|f^*\|_{L^2(\mu)}^2}{n^2} + \frac{n}{D} (\|f^*\|_{L^2(\mu)}^2 + \sigma^2) \right). \quad (9.12)$$

There are a number of observations to be made about this result. First, to control the error probability, it is not enough to choose δ small and n large. One also needs to choose D to be large

relative to n^2 . Second, the dependence of the error bound on δ may look unexpectedly weak. This is the result of deliberately loose analysis: The present bound will be sufficient for our purposes (see [Corollary 9.3.3](#)) and the analysis can be shorter this way. The third observation is that the error bound in [Eq. \(9.12\)](#) has familiar and less familiar terms. In particular, the first two terms are familiar from standard analysis of regression: the approximation and estimation errors need to be balanced to get good outcomes. If we used m coefficients with standard linear least squares, these are the terms one would expect to see. The next, third term, is a small shrinkage bias. The last term is unusual: This is the price of fitting to the residual error (after the first m coefficients are used to get the “signal”). To make this error small, D needs to be chosen large relative to n (similarly to how D needs to be large to control the error probability). For a fixed n , to get the best possible error bound, first one could choose a value of m to balance between the first two terms (while also paying attention to the error probability term), after which the value of D is dictated by the need to control the last term (and the error probability term). Increasing D supplies more directions over which to distribute the interpolation correction. Surprisingly, increasing D has no price attached to it – other than the increased computation and memory costs. This is because of the careful selection of the weighting of the Chebyshev polynomial basis functions: As D gets large, these are appropriately downweighted and as such, the upper bound on the squared error of the resulting residual component decreases as D increases. In effect, the procedure closely follows what the point-correction mechanism is aimed to achieve: The function that is learnt with the last D coefficients approximates the point-mass correction, while the function that is learnt with the first m coefficients approximates the function that would be learnt using just those basis functions.

Again, the intuitive explanation is that when solving for the minimum norm interpolant, the price for the last D coefficients is much higher than for the first m . The result of this is essentially a two-step fitting: First fit using the first m coefficients, then use the last D coefficients to approximate the residual. [Exercise 7](#) asks you to make this comparison precise and bound the difference between the coefficient vectors. In general, the first stage is generalized least squares, with residual weight $(VV^\top)^{-1}$ for the second block’s sample matrix V , defined next. It reduces to ordinary least squares when $VV^\top$ is a scalar multiple of I_n .

Define

$$U_{ij} = \psi_j(X_i) \quad (0 \leq j < m), \quad V_{ij} = \chi_{m+j}(X_i) \quad (0 \leq j < D), \quad H = D^{-1}VV^\top.$$

Let $u = (\psi_0, \dots, \psi_{m-1})^\top$ and $v = (\chi_m, \dots, \chi_{m+D-1})^\top$. For matrices, $\|\cdot\|$ denotes the Euclidean operator norm, and $S \preceq R$ means that $R - S$ is positive semidefinite. In [Appendix 9.B](#), it is shown that for universal constants $c_H, C_H, c_{\text{pr}}, C_{\text{pr}} > 0$, there is an event $\mathcal{G}$ determined by the inputs such that

$$c_H I_n \preceq H \preceq C_H I_n, \quad U^\top U \succeq (n/2) I_m \quad \text{while} \quad \mathbb{P}(\mathcal{G}^c) \leq C_{\text{pr}} n^2 / D + m \exp(-c_{\text{pr}} n / m^2). \quad (9.13)$$

The second condition makes the low-degree coefficients estimable. To see the role of the first, fix any vector $z \in \mathbb{R}^n$, meant to model the “residual errors” after removing the “signal”. On $\mathcal{G}$, the coefficients $b_z = V^\top (VV^\top)^{-1} z$ give $Vb_z = z$, while

$$\|b_z^\top v\|_{L^2(\mu)}^2 \leq \frac{\pi}{2} \|b_z\|^2 = \frac{\pi}{2} z^\top (VV^\top)^{-1} z \leq \frac{\pi}{2c_H D} \|z\|^2. \quad (9.14)$$

The first inequality is due to [Eq. \(9.9\)](#), while the last is from $VV^\top = DH \succeq c_H D I_n$. A residual vector with squared length of order n can therefore be fitted at population cost of order n/D . This is the interpolation mechanism. It uses a population-norm bound for the second block and a well-conditioned sample Gram matrix.

Proof of Theorem 9.3.1. Let $\mathcal{G}$ be the input event supplied by Eq. (9.13). As noted above, Appendix 9.B proves Eq. (9.13). Let $\Delta := \|\hat{f} - f^*\|_{L^2(\mu)}^2$. Now, for any $s \geq 0$,

$$\mathbb{P}(\Delta > s) \leq \mathbb{P}(\Delta > s, \mathcal{G}) + \mathbb{P}(\mathcal{G}^c) = \mathbb{P}(\mathbb{I}\{\mathcal{G}\}\Delta > s) + \mathbb{P}(\mathcal{G}^c),$$

where the last equality holds because $\{\Delta > s\} \cap \mathcal{G} = \{\mathbb{I}\{\mathcal{G}\}\Delta > s\}$. Indeed, if $\mathcal{G}$ holds and $\Delta > s$, $\mathbb{I}\{\mathcal{G}\} = 1$ and thus $\mathbb{I}\{\mathcal{G}\}\Delta > s$ also holds and if $\mathbb{I}\{\mathcal{G}\}\Delta > s$ then $\mathbb{I}\{\mathcal{G}\} = 1$ must be true (since s is nonnegative), hence $\mathcal{G}$ holds and also $\Delta > s$ holds. Now, choosing $s = \frac{1}{\delta} \mathbb{E}[\mathbb{I}\{\mathcal{G}\}\Delta]$, from Markov's inequality we get

$$\mathbb{P}(\Delta > \frac{1}{\delta} \mathbb{E}[\mathbb{I}\{\mathcal{G}\}\Delta]) \leq \delta + \mathbb{P}(\mathcal{G}^c).$$

If $\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\Delta] = 0$, the same conclusion follows because $\mathbb{I}\{\mathcal{G}\}\Delta = 0$ almost surely. Hence, it suffices to bound $\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\Delta]$ by C times the expression in parentheses in Eq. (9.12).

Define the two coefficient vectors together by

$$(\hat{\beta}, \hat{b}) = \arg \min_{\beta \in \mathbb{R}^m, b \in \mathbb{R}^D} \{ \|\beta\|^2 + D\|b\|^2 : \mathbf{Y} = U\beta + Vb \}.$$

Let

$$g_m = \beta_\star^\top u = \arg \min_{g \in \mathcal{F}_m} \|g - f^*\|_{L^2(\mu)}^2$$

be the best degree- $(m-1)$ polynomial approximation of f^* in $L^2(\mu)$, and put $r = f^* - g_m$. Then

$$\|r\|_{L^2(\mu)}^2 = A_m(f^*), \quad \|\beta_\star\|^2 = \|g_m\|_{L^2(\mu)}^2 \leq \|g_m\|_{L^2(\mu)}^2 + \|r\|_{L^2(\mu)}^2 = \|f^*\|_{L^2(\mu)}^2. \quad (9.15)$$

Here, $\|\beta_\star\|^2 = \|g_m\|_{L^2(\mu)}^2$ is due to the choice of the orthonormal basis u , and the last equality is because the optimality of g_m makes r orthogonal to g_m . Exercise 5 asks you to prove this orthogonality and the resulting Pythagorean identity.

Set $h = \hat{b}^\top v$. Since

$$\hat{f} - f^* = (\hat{\beta} - \beta_\star)^\top u + h - r$$

from the triangle inequality, $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$, and because the components of u are orthonormal in $L^2(\mu)$, we get

$$\Delta = \|\hat{f} - f^*\|_{L^2(\mu)}^2 \leq 3\|\hat{\beta} - \beta_\star\|^2 + 3\|h\|_{L^2(\mu)}^2 + 3A_m(f^*). \quad (9.16)$$

(Since the feature groups are not orthogonal in $L^2(\mu)$, the triangle inequality is the best tool we have.)

It remains to bound $\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\|\hat{\beta} - \beta_\star\|^2]$ and $\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\|h\|_{L^2(\mu)}^2]$. To bound these, the next step is to calculate $(\hat{\beta}, \hat{b})$ in closed form. For this, we first separate the minimization over the two coefficient blocks:

$$\begin{aligned} & \min \{ \|\beta\|^2 + D\|b\|^2 : \mathbf{Y} = U\beta + Vb, \beta \in \mathbb{R}^m, b \in \mathbb{R}^D \} \\ &= \min \{ \min \{ \|\beta\|^2 + D\|b\|^2 : \mathbf{Y} = U\beta + Vb, b \in \mathbb{R}^D \} : \beta \in \mathbb{R}^m \} \\ &= \min \{ \|\beta\|^2 + \min \{ D\|b\|^2 : \mathbf{Y} = U\beta + Vb, b \in \mathbb{R}^D \} : \beta \in \mathbb{R}^m \}. \end{aligned}$$

For fixed β , the minimum-Euclidean-norm solution of $Vb = \mathbf{Y} - U\beta$ on $\mathcal{G}$ is

$$b_\beta = V^\top (VV^\top)^{-1} (\mathbf{Y} - U\beta). \quad (9.17)$$

By algebra, we get $D\|b_\beta\|^2 = (\mathbf{Y} - U\beta)^\top H^{-1}(\mathbf{Y} - U\beta)$. Plugging this value into the outer problem reduces it to minimizing $\|\beta\|^2 + (\mathbf{Y} - U\beta)^\top H^{-1}(\mathbf{Y} - U\beta)$. Setting the derivative of this with respect to β to zero and solving gives

$$\widehat{\beta} = (I_m + U^\top H^{-1}U)^{-1}U^\top H^{-1}\mathbf{Y}. \quad (9.18)$$

Exercise 3 asks you to supply the linear algebra and differentiation steps in this elimination.

Write $r_X = (r(X_i))_{i=1}^n$. Then, from $f^\star = g_m + f^\star - g_m = \beta_\star^\top u + r$ and $Y_i = f^\star(X_i) + \varepsilon_i$ we get

$$\mathbf{Y} = U\beta_\star + r_X + \varepsilon. \quad (9.19)$$

On $\mathcal{G}$, define

$$A = U^\top H^{-1}U, \quad T = (I_m + A)^{-1}U^\top H^{-1},$$

and set $A = 0$ and $T = 0$ on $\mathcal{G}^c$ when taking expectations below. On $\mathcal{G}$, Eq. (9.18) becomes $\widehat{\beta} = T\mathbf{Y}$, which combined with Eq. (9.19) gives

$$\widehat{\beta} - \beta_\star = -(I_m + A)^{-1}\beta_\star + Tr_X + T\varepsilon, \quad (9.20)$$

thus,

$$\begin{aligned} \|\widehat{\beta} - \beta_\star\|^2 &\leq 3(\|(I_m + A)^{-1}\|^2 \|\beta_\star\|^2 + \|Tr_X\|^2 + \|T\varepsilon\|^2) \\ &\leq 3(\|(I_m + A)^{-1}\|^2 \|f^\star\|_{L^2(\mu)}^2 + \|Tr_X\|^2 + \|T\varepsilon\|^2), \end{aligned}$$

where the last inequality used that by Eq. (9.15), $\|\beta_\star\|^2 \leq \|f^\star\|_{L^2(\mu)}^2$.

Since we need to bound $\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\|\widehat{\beta} - \beta_\star\|^2]$, we turn to bounding the expectation of the three terms multiplied by the indicator of $\mathcal{G}$.

We have $\|Tr_X\|^2 = r_X^\top T^\top Tr_X \leq \|T^\top T\| \|r_X\|^2$. Thus, if B_T is a deterministic upper bound on $\|T^\top T\|$ on $\mathcal{G}$,

$$\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\|Tr_X\|^2] \leq B_T \mathbb{E}[\|r_X\|^2] = B_T n \mathbb{E}[r^2(X_1)] = B_T n \|r\|_{L^2(\mu)}^2 = B_T n A_m(f^\star).$$

Next, defining $X_{1:n} = (X_1, \dots, X_n)$, we calculate $\|T\varepsilon\|^2 = \varepsilon^\top T^\top T\varepsilon = \text{tr}(T^\top T\varepsilon\varepsilon^\top)$. Thus, $\mathbb{E}[\|T\varepsilon\|^2 | X_{1:n}] = \text{tr}(T^\top T\mathbb{E}[\varepsilon\varepsilon^\top | X_{1:n}]) \leq \sigma^2 \text{tr}(T^\top T I_n) = \sigma^2 \text{tr}(T^\top T)$, where the i.i.d. sampling assumption and Eq. (9.6) give $\mathbb{E}[\varepsilon | X_{1:n}] = 0$ and $\mathbb{E}[\varepsilon\varepsilon^\top | X_{1:n}] \preceq \sigma^2 I_n$. Thus,

$$\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\|T\varepsilon\|^2] = \mathbb{E}[\mathbb{E}[\mathbb{I}\{\mathcal{G}\}\|T\varepsilon\|^2 | X_{1:n}]] \leq \sigma^2 \mathbb{E}[\mathbb{I}\{\mathcal{G}\} \text{tr}(TT^\top)].$$

Thus, it remains to bound $TT^\top$ and $\|(I_m + A)^{-1}\|$ on $\mathcal{G}$. On $\mathcal{G}$, the two sides of the bound on H in Eq. (9.13) give

$$H^{-1} \succeq C_H^{-1} I_n, \quad H^{-2} \preceq c_H^{-1} H^{-1}.$$

Together with $U^\top U \succeq (n/2)I_m$, the first inequality yields

$$A = U^\top H^{-1}U \succeq C_H^{-1}U^\top U \succeq \frac{n}{2C_H}I_m.$$

Hence

$$\begin{aligned} \|(I_m + A)^{-1}\| &\leq \frac{1}{1 + n/(2C_H)} \leq \frac{2C_H}{n}, \\ TT^\top &\preceq c_H^{-1}(I_m + A)^{-1}A(I_m + A)^{-1} \preceq \frac{2C_H}{c_H n}I_m. \end{aligned} \quad (9.21)$$

From the last inequality and $\|T^\top T\| = \|TT^\top\|$, we see that we can choose $B_T = \frac{2C_H}{c_H n}$, and we also have $\text{tr}(TT^\top) \leq \frac{2C_H m}{c_H n}$. [Exercises 9](#) and [10](#) ask you to justify these matrix-order, norm, and trace steps, including the conditional noise calculation. Putting things together, we get

$$\mathbb{E}[\|\mathcal{G}\|\hat{\beta} - \beta_\star\|^2] \leq C \left(\frac{\|f^\star\|_{L^2(\mu)}^2}{n^2} + A_m(f^\star) + \frac{\sigma^2 m}{n} \right) \quad (9.22)$$

for some universal constant $C > 0$.

We now turn to bounding $\|h\|_{L^2(\mu)}^2 = \|\hat{b}^\top v\|_{L^2(\mu)}^2$ in [Eq. \(9.16\)](#). By [Eq. \(9.9\)](#), $\|\hat{b}^\top v\|_{L^2(\mu)}^2 \leq \frac{\pi}{2} \|\hat{b}\|^2$, so it remains to bound $\|\hat{b}\|^2$. For this, set

$$K = UU^\top + H, \quad \alpha = K^{-1}\mathbf{Y}.$$

[Exercise 8](#) asks you to derive the following identities from the optimality equations and to verify them directly from [Eqs. \(9.17\)](#) and [\(9.18\)](#) using the push-through identity:

$$\hat{\beta} = U^\top \alpha, \quad \hat{b} = D^{-1}V^\top \alpha.$$

Thus $VV^\top = DH$ and $c_H I_n \preceq H \preceq K$ give

$$\|\hat{b}\|^2 = \frac{1}{D} \alpha^\top H \alpha \leq \frac{1}{D} \mathbf{Y}^\top K^{-1} \mathbf{Y} \leq \frac{1}{c_H D} \|\mathbf{Y}\|^2. \quad (9.23)$$

Since $\mathbb{E}[\|\mathbf{Y}\|^2] = n\mathbb{E}[Y_1^2] \leq n(\|f^\star\|_{L^2(\mu)}^2 + \sigma^2)$, it follows that

$$\mathbb{E}[\|\mathcal{G}\|\|h\|_{L^2(\mu)}^2] \leq \frac{\pi n}{2c_H D} (\|f^\star\|_{L^2(\mu)}^2 + \sigma^2).$$

Combining this bound with [Eqs. \(9.16\)](#) and [\(9.22\)](#) gives the required bound on $\mathbb{E}[\|\mathcal{G}\|\Delta]$. The initial Markov argument and the bound on $\mathbb{P}(\mathcal{G}^c)$ then prove the theorem, after increasing C to cover both the risk bound and C_{pr} , and taking $c = c_{\text{pr}}$. $\square$

Why use Chebyshev polynomials in the second block? The proof uses their population-norm bound [Eq. \(9.9\)](#) and the bounds on H in [Eq. \(9.13\)](#). Legendre polynomials would also give the population-norm bound, with equality. The advantage of Chebyshev polynomials is in controlling H uniformly in the block of size D . The boundedness and cosine representation of these polynomials imply that

$$H_{ii} = \frac{1}{D} \sum_{j=m}^{m+D-1} \chi_j(X_i)^2$$

is bounded above and away from zero by universal constants when $D \geq 16m$. Their orthonormality under ν , together with $\mu \leq (\pi/2)\nu$, also bounds the off-diagonal second moments: $\mathbb{E}[H_{i\ell}^2] \leq \pi^2/(4D)$ for $i \neq \ell$. In [Appendix 9.B](#), these facts give both of the high probability spectral bounds on H . The upper bound on H controls the estimation error through $A = U^\top H^{-1}U$; the lower bound controls the coefficient cost of fitting the residual in [Eq. \(9.23\)](#).

For comparison, using the normalized Legendre polynomials in the second block would give, at $x = 1$,

$$\frac{1}{D} \sum_{j=m}^{m+D-1} \psi_j(1)^2 = \frac{1}{D} \sum_{j=m}^{m+D-1} (2j+1) = 2m + D.$$

Thus the uniform diagonal upper bound used in this argument would fail. Chebyshev polynomials supply that bound while retaining the population-norm control needed for the interpolation component. This is not merely a limitation of the proof: replacing the Chebyshev block by Legendre polynomials while retaining the same scaling can destroy consistency (see next section), even for a bounded continuous target without observation noise.⁵

9.3.3 Consistency and rates

Does the construction give a genuinely nontrivial benign set? As it turns out for our method the benign set cannot be further from being trivial:

Corollary 9.3.2 (Universal consistency). *Let B^{poly} use the feature map in Eq. (9.10) with*

$$m_n = \max\{1, \lfloor n^{1/4} \rfloor\}, \quad D_n = n^3,$$

and select the minimum-Euclidean-norm interpolant. With these choices, every law satisfying Eq. (9.6) with $f^ \in L^2(\mu)$ belongs to $\mathcal{A}(B^{\text{poly}})$.*

We call this a universal consistency result. “Universal” here refers to that the only restriction on the regression function f^* is that it needs to be square integrable with respect to μ (which is necessary anyways for the risk to be well-defined). The law otherwise is a bit more restricted than what one usually sees in standard “universal consistency” results in that we assumed a uniform bound on the noise variance. We note in passing that with extra work, the results extend to the case when only the existence of the expected risk is assumed. In particular, [Exercise 11](#) asks you to prove this extension. The uniform conditional variance bound was responsible for the explicit $\sigma^2 m/n$ term; the exercise asks you to replace that term by a bound depending on the tail of the conditional variance.

Proof. Polynomials are dense in $L^2(\mu)$, so $A_{m_n}(f^*) \rightarrow 0$. Fix $\eta > 0$ and apply [Theorem 9.3.1](#) with $\delta = \eta/2$. Every term on the right-hand side of [Eq. \(9.12\)](#) tends to zero, and the remaining exceptional-event probability tends to zero. Hence $\|B_n^{\text{poly}}(S_n) - f^*\|_{L^2(\mu)}^2 \xrightarrow{\mathbb{P}} 0$ as $n \rightarrow \infty$. $\square$

What rate is available when the target has additional structure? The target enters the bound through $A_m(f^*)$, while m/n is the cost of estimating the first m coefficients from noisy observations. Recall the polynomial approximation condition [Eq. \(9.8\)](#) and the smoothness examples given there. We next use these to derive that the minimum-norm interpolant is not only universally consistent, but also enjoys the best possible nonparametric rates for smooth functions.

To state this result we recall the definition of the $O_{\mathbb{P}}(\cdot)$ notation: For real random variables Z_n and deterministic numbers $a_n > 0$, the notation $Z_n = O_{\mathbb{P}}(a_n)$ means that Z_n/a_n is *bounded in probability*: for every $\delta > 0$, there are a constant $0 < C_\delta < \infty$ and an integer n_δ such that

$$\mathbb{P}(|Z_n| > C_\delta a_n) \leq \delta \quad \text{for all } n \geq n_\delta.$$

If $a_n \rightarrow 0$, this implies $Z_n \xrightarrow{\mathbb{P}} 0$. (However, note that no bound on $\mathbb{E}[|Z_n|]$ follows from this.)

Corollary 9.3.3 (s -smooth targets). *Fix $s > 1/2$ and choose*

$$m_n = \max\{1, \lfloor n^{1/(2s+1)} \rfloor\}, \quad D_n = n^3.$$

For every fixed target satisfying Eq. (9.8) and noise law satisfying Eq. (9.6),

$$\|\hat{f}_n - f^*\|_{L^2(\mu)}^2 = O_{\mathbb{P}}(n^{-2s/(2s+1)}) \longrightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (9.24)$$

Thus these noisy problems belong to $\mathcal{A}(B_s)$ for one predetermined procedure B_s . For Lipschitz targets the rate is $O_{\mathbb{P}}(n^{-2/3})$.

Proof. The choices balance $A_{m_n}(f^*) = O(m_n^{-2s})$ with m_n/n at order $n^{-2s/(2s+1)}$. The shrinkage and interpolation terms are $O(n^{-2})$. Also $D_n \geq 16m_n$ eventually, $n/m_n^2 \asymp n^{(2s-1)/(2s+1)}$ grows polynomially, and $n^2/D_n = n^{-1}$. For any fixed error probability, choose δ smaller than that probability. The other exceptional-event terms tend to zero, giving the asserted bound in probability. $\square$

Choosing m_n this way presumes the approximation class. For each chosen s , the method is fixed before seeing the target or sample. The exponent $2s/(2s+1)$ is the classical optimal exponent for one-dimensional noisy regression on bounded s -smooth classes. Here it is attained in probability.

The restriction $s > 1/2$ ensures $m_n^2 \log m_n = o(n)$, which our Legendre-design concentration bound requires. This is a restriction of the theorem, not an impossibility result for rougher targets.⁶ The proof uses the sufficient schedule $D_n = n^3$. The important separation is that m_n balances approximation and estimation, whereas the much larger D_n makes interpolation inexpensive in population loss.

Does convergence in probability also give convergence of expected risk? We have already mentioned that convergence in probability does not imply convergence in expectation. As it happens, for the unmodified polynomial interpolator, the expected risk is in fact infinite at each $n \geq 2$! [Appendix 9.C](#) ask you to prove this.

The mechanism of failure is instructive. Our procedure is such that it chooses smooth functions. As an example, assume that for some $C > 0$, for any f chosen by the method and any $x_1, x_2 \in [-1, 1]$, $|f(x_1) - f(x_2)| \leq C|x_1 - x_2| \|f\|_{L^2(\mu)}$. The squared norm on the right-hand side can appear because $\mathcal{F}$ in our case is a finite-dimensional space, and as such, any norm can be replaced by any other norm, at the price of changing constants. Since the chosen function $\hat{f}$ needs to interpolate data, it may happen that for $x_1 = X_1$, $x_2 = X_2$, $|Y_1 - Y_2| = |f(X_1) - \hat{f}(X_2)|$ is large, while X_1 and X_2 are close to each other. But the above inequality must hold for $\hat{f}$, which means that $\|\hat{f}\|_{L^2(\mu)}$ must be large, which will be a problem when f^* has a small squared norm! That this mechanism causes the expected risk of the method to explode means that the probability of seeing large errors is poorly controlled; the procedure has high risk.

This, of course, is not a desired property. Luckily, at least in the case when the responses Y are known to belong to some known interval $[-M, M]$, the issue can be addressed by simply truncating predictions to $[-M, M]$. The resulting method will still interpolate, and its error can only be smaller than that of the original method. In fact, one can show that this clipped method now gives the same rate that we calculated above but also in expected risk.

9.3.4 The same polynomial space with another norm

Does the same polynomial space work with any norm? To test the role of the norm, keep the input law, target, noise, and polynomial space unchanged. Express the entire polynomial in the orthonormal Legendre basis and give all coefficients equal penalty. This selects the interpolating polynomial of smallest $L^2(\mu)$ norm.

Proposition 9.3.4 (Equal Legendre coefficient penalties can discard the signal). *Use m_n, D_n from [Corollary 9.3.3](#), and let $\hat{f}_n^{\text{eq}}$ minimize $\|f\|_{L^2(\mu)}$ among interpolating polynomials of degree less than $m_n + D_n$. For every fixed $f^* \in L^2(\mu)$ under [Eq. \(9.6\)](#),*

$$\|\hat{f}_n^{\text{eq}}\|_{L^2(\mu)}^2 = O_{\mathbb{P}}(n/D_n) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Consequently,

$$\|\hat{f}_n^{\text{eq}} - f^*\|_{L^2(\mu)}^2 \xrightarrow{\mathbb{P}} \|f^*\|_{L^2(\mu)}^2 \quad \text{as } n \rightarrow \infty.$$

Thus every fixed nonzero target defines a problem outside the benign set of this equal-penalty procedure.

Proof. On the event $H \succeq cI_n$, the polynomial $h_0(x) = v(x)^\top V^\top (VV^\top)^{-1} \mathbf{Y}$ interpolates. By minimality and Eq. (9.9),

$$\|\widehat{f}_n^{\text{eq}}\|_2^2 \leq \|h_0\|_2^2 \leq \frac{\pi}{2} \mathbf{Y}^\top (VV^\top)^{-1} \mathbf{Y} \leq \frac{C}{D} \|\mathbf{Y}\|^2.$$

Since $\|\mathbf{Y}\|^2 = O_{\mathbb{P}}(n)$ and this event has probability tending to one, the first claim follows. Expand the squared distance and apply Cauchy–Schwarz to obtain the second. $\square$

Minimizing the equal-coefficient norm therefore selects interpolants whose $L^2(\mu)$ norm tends to zero, even when the target is nonzero. The positive rule instead makes the low-degree signal inexpensive relative to fitting all responses with the additional features. Both rules interpolate, and their population limits differ. We next ask whether commonly used kernel norms provide the needed bias.

9.4 What does “minimum norm” mean?*

Kernel methods strictly generalize the finite-dimensional linear approach of Section 9.3: every finite feature model has a kernel representation, but a kernel can also represent an infinite-dimensional feature space. Working with infinitely many features is appealing because it can provide greater approximation power, while the fitted predictor can still be computed using a finite kernel matrix. It is therefore natural to ask whether the ideas behind our polynomial construction extend to this setting, and how: can minimum-norm interpolation use this flexibility to learn the signal while fitting the noise at vanishing population cost?

We first connect the kernel terminology to the linear models of Section 9.3.

From linear features to kernels. A feature map $\varphi : \mathcal{X} \rightarrow \mathbb{R}^p$ gives both a linear function class and a function recording inner products between feature vectors:

$$f_\theta(x) = \theta^\top \varphi(x), \quad k(x, x') = \varphi(x)^\top \varphi(x').$$

The model is linear in θ , although its features may be nonlinear in x . The function k is symmetric, and every finite matrix of its values is positive semidefinite, since

$$\sum_{i,j=1}^N c_i c_j k(x_i, x_j) = \left\| \sum_{i=1}^N c_i \varphi(x_i) \right\|^2 \geq 0.$$

A symmetric real-valued function with this property for all N , inputs $x_1, \dots, x_N$, and real coefficients $c_1, \dots, c_N$ is called a *positive semidefinite kernel*. This is different from a probability kernel: it records inner products, not conditional distributions.

For orientation, familiar kernels on $\mathbb{R}^d$ include the linear kernel $k(x, x') = x^\top x'$ and the polynomial kernel $k(x, x') = (1 + x^\top x')^q$, where q is a positive integer. These have finite-dimensional feature representations. Two standard infinite-dimensional examples are the Laplace and Gaussian kernels:

$$k_\gamma^{\text{Lap}}(x, x') = \exp(-\|x - x'\|/\gamma), \quad k_\gamma^{\text{Gauss}}(x, x') = \exp(-\|x - x'\|^2/(2\gamma^2)).$$

Both are *radial*: their values depend only on the Euclidean distance between the inputs. The parameter $\gamma > 0$ is the *bandwidth*, or length scale. Smaller bandwidths make the kernel more localized around $x = x'$; larger bandwidths make it decay more slowly with distance.

The associated function space. For these finite features, let $\mathcal{H}_k = \{f_\theta : \theta \in \mathbb{R}^p\}$ and give it the induced norm $\|f\|_\varphi$ from [Section 9.3](#). Its inner product is the Euclidean inner product of the minimum-norm coefficient representatives. Thus linearly dependent features cause no ambiguity. The function $k(x, \cdot)$ belongs to this space and represents evaluation:

$$f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}_k}.$$

This is the *reproducing property*. More generally, a *reproducing kernel Hilbert space* (RKHS) is a Hilbert space of real-valued functions on $\mathcal{X}$ in which each point evaluation is continuous. Equivalently, each evaluation has a representative $k(x, \cdot)$ satisfying the displayed identity. In particular, Cauchy–Schwarz gives

$$|f(x)| \leq \|f\|_{\mathcal{H}_k} \sqrt{k(x, x)}.$$

An RKHS therefore controls point values, unlike $L^2(\mu)$ for a point-mass-free μ . The feature space may be infinite-dimensional: taking $\varphi(x) = k(x, \cdot) \in \mathcal{H}_k$ gives $k(x, x') = \langle \varphi(x), \varphi(x') \rangle_{\mathcal{H}_k}$ and $f(x) = \langle f, \varphi(x) \rangle_{\mathcal{H}_k}$. Kernel prediction is still linear function approximation in this feature space.

Feature scaling determines the norm. To compare the RKHS norm with prediction error, we also need the test-input law μ . For a bounded measurable kernel k , define its *integral operator* $T_{k,\mu} : L^2(\mu) \rightarrow L^2(\mu)$ by

$$(T_{k,\mu}g)(x) = \int_{\mathcal{X}} k(x, x')g(x')\mu(dx'). \quad (9.25)$$

This is the population analogue of multiplying a vector of function values by a kernel matrix: the integral replaces a finite weighted sum. Its eigenfunctions identify orthogonal directions in $L^2(\mu)$ in which the RKHS norm has a simple coefficient penalty, as we now explain.

Suppose the kernel has the pointwise expansion

$$k(x, x') = \sum_{j \geq 0} \lambda_j \psi_j(x) \psi_j(x'), \quad \lambda_j > 0, \quad (9.26)$$

with the functions ψ_j orthonormal in $L^2(\mu)$. Substituting into [Eq. \(9.25\)](#) and using orthonormality gives

$$T_{k,\mu}\psi_j = \lambda_j \psi_j.$$

Thus the λ_j and ψ_j are eigenvalues and eigenfunctions of this operator, which depends on both the kernel and the input law. The corresponding features are $\sqrt{\lambda_j} \psi_j(x)$. Representing $f = \sum_j a_j \psi_j \in \mathcal{H}_k$ therefore uses coefficients $a_j / \sqrt{\lambda_j}$, giving

$$\|f\|_{L^2(\mu)}^2 = \sum_{j \geq 0} a_j^2, \quad \|f\|_{\mathcal{H}_k}^2 = \sum_{j \geq 0} \frac{a_j^2}{\lambda_j}. \quad (9.27)$$

The significance of the eigenvalues is this comparison: a component $a_j \psi_j$ has squared population norm a_j^2 , but incurs squared RKHS norm a_j^2 / λ_j . Larger eigenvalues therefore mean a smaller penalty for a component of the same population size. The operator makes the relation between the norm minimized by the algorithm and the norm used to measure prediction error explicit. Thus a minimum-norm claim is incomplete until the feature map and its scaling, or equivalently the kernel, have been specified. The weighted polynomial rule uses the finite kernel

$$k_{m,D}(x, x') = \sum_{j < m} \psi_j(x) \psi_j(x') + \frac{1}{D} \sum_{m \leq j < m+D} \chi_j(x) \chi_j(x'). \quad (9.28)$$

For $f_{\beta,b}(x) = u(x)^\top \beta + v(x)^\top b$, the squared RKHS norm is

$$\|f_{\beta,b}\|_{\mathcal{H}_{k_m,D}}^2 = \|\beta\|^2 + D\|b\|^2.$$

Because the combined Legendre–Chebyshev basis is not orthogonal under μ , these coefficient weights are not the reciprocals of the eigenvalues of $T_{k_m,D,\mu}$. They would be for an orthonormal basis, as in Eq. (9.27). The larger penalty makes each additional coefficient expensive, while the number of features makes collective interpolation inexpensive in population norm. Exercise 6 asks you to verify the orthonormal case.

Choosing a kernel. A useful kernel admits good approximations to plausible targets at a small RKHS norm. For regression, the relevant smoothness is that of the conditional mean f^* , not of the noisy responses. Among standard radial kernels at comparable bandwidths, smoother choices penalize fine-scale variation more strongly. In regularized regression this can improve estimation when the target is smooth. A rougher kernel may approximate corners or less regular behavior with a smaller norm, but also allows more freedom to fit noise. Thus kernel choice involves both approximation and estimation; “smoother” does not mean “better.” The target need not belong to the RKHS: what matters is how well it can be approximated at a controlled norm.

The *Matérn family* offers a range of smoothness choices. Its parameter $\nu > 0$ controls smoothness separately from its bandwidth γ : $\nu = 1/2$ gives the Laplace kernel, larger ν gives smoother kernels, and the Gaussian kernel is the limit as $\nu \rightarrow \infty$ with γ fixed. Smoothness and bandwidth are different: a Gaussian kernel with a small bandwidth is still infinitely differentiable, but can represent variation on short distance scales.

In practice, choose the features, distance scale, and kernel family using knowledge of the problem, then compare bandwidths and smoothness choices using held-out prediction error or cross-validation. With noisy data, one commonly also tunes a regularization penalty: *kernel ridge regression* adds $\lambda\|f\|_{\mathcal{H}_k}^2$, with $\lambda > 0$, to the average squared training loss. This permits residuals instead of requiring the fit to reproduce every noisy response. These considerations guide kernel selection, but do not by themselves establish that minimum-norm interpolation is consistent.

For a fixed RKHS $\mathcal{H}_k$, the standard ridgeless estimator is

$$\hat{f}_k \in \arg \min_{f \in \mathcal{H}_k} \|f\|_{\mathcal{H}_k} \quad \text{subject to} \quad f(X_i) = Y_i \text{ for every nonconflicting index } i. \quad (9.29)$$

Under the point-mass-free laws studied below, all indices are nonconflicting almost surely, and these are the usual n interpolation constraints. The minimum-norm solution lies in the span of $k(X_1, \cdot), \dots, k(X_n, \cdot)$: an orthogonal component vanishes at all sample inputs by the reproducing property and only increases the norm. This is the interpolation form of the so-called *representer theorem*. Writing the solution as $\sum_i \alpha_i k(X_i, \cdot)$ turns the constraints into $K\alpha = \mathbf{Y}$, where $K = (k(X_i, X_j))_{i,j=1}^n$. When K is invertible, define $k(x, \mathbf{X}) = (k(x, X_1), \dots, k(x, X_n))^\top$ to obtain

$$\hat{f}_k(x) = k(x, \mathbf{X})^\top K^{-1} \mathbf{Y}. \quad (9.30)$$

This is also the zero-ridge limit of kernel ridge regression and, under the corresponding linearized dynamics, the endpoint selected by gradient flow. These facts make Eq. (9.29) a standard proposal. The next two results ask when its selected interpolant is consistent.

9.4.1 A fixed-problem failure for the Laplace kernel

Does a conventional fixed kernel already have the needed norm, perhaps after tuning its bandwidth? The following fixed noisy problem gives a negative answer for the Laplace family. This is a standard

kernel with substantial approximation power: on a compact domain, its RKHS can approximate every continuous function uniformly. Its failure below therefore concerns the minimum-norm fitting rule, not a lack of expressive power.

Let μ be uniform on the unit ball in $\mathbb{R}^3$.⁷ Take

$$Y = 1 + \varepsilon, \quad \mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2},$$

with ε independent of X , and use the Laplace kernels

$$k_c(x, x') = c^3 \exp(-c\|x - x'\|), \quad c > 0.$$

Here c is the inverse bandwidth: the bandwidth is $1/c$. The positive factor c^3 rescales the RKHS norm but does not change its minimizing interpolant for a fixed c . For each c , let $\hat{f}_{n,c}$ be the minimum-RKHS-norm interpolant.

Theorem 9.4.1 (Laplace minimum-norm interpolation is inconsistent). *There are constants $a, C > 0$ such that, for all sufficiently large n ,*

$$\mathbb{P} \left(\inf_{c>0} \|\hat{f}_{n,c} - 1\|_{L^2(\mu)}^2 \geq a \right) \geq 1 - \frac{C}{\sqrt{n}}. \quad (9.31)$$

The statement holds simultaneously for every $c > 0$, so c may be selected from the data.

The result places one fixed, elementary noisy problem outside the benign set of every Laplace-bandwidth selection rule. It is stronger for our purpose than a minimax statement whose difficult distribution may change with n . Its interest comes from the procedure: minimum RKHS norm is not an intentionally poor interpolation rule.

This result complements the scaled all-Legendre counterexample in [Endnote 5](#). Both show that a rich function class and minimum-norm selection do not suffice for consistency. The failure mechanisms differ. In the all-Legendre example, the induced weighted approximation can fail in the population L^2 norm even for a bounded continuous target without noise. Here, interpolation of noisy observations prevents consistency even for a constant target. Moreover, the Laplace result covers every bandwidth choice, whereas the all-Legendre counterexample concerns the particular scaling and dimension schedules considered there. Whether tuning can restore universal consistency for the all-Legendre family remains open here.⁸

The proof has two regimes. When c is not too large, fitting noise at nearby sample points forces enough variation to create nonvanishing integrated error. When c is large, the minimum-norm interpolant is concentrated near the observations and has too little L^2 mass to approximate the nonzero regression function. Every bandwidth falls into one of these regimes in fixed dimension. The theorem concerns this kernel family and this fixed-dimensional setting.

9.4.2 A published positive counterpart

Can a different kernel reproduce the separation that worked for the series estimator? A positive theorem uses a norm designed for exactly that purpose. Let $\tilde{k}$ be a continuous universal kernel on a compact domain $\mathcal{X} \subset \mathbb{R}^d$: its RKHS is dense in the continuous functions in the uniform norm. A Gaussian kernel with any fixed positive bandwidth is one such choice of $\tilde{k}$. Let $\check{k}_\gamma(x, x') = \exp(-\|x - x'\|/\gamma)$ be a narrow Laplace kernel. Define the sample-size-dependent kernel

$$k_n = \tilde{k} + \rho_n \check{k}_{\gamma_n}. \quad (9.32)$$

Theorem 9.4.2 (Spiky-smooth minimum-norm interpolation). *Suppose $P_X(B_t(x)) \leq C_X t^{\beta_X}$ for all $x \in \mathcal{X}$ and $0 < t \leq 1$, for constants $C_X, \beta_X > 0$, and $\mathbb{E}[Y^2] < \infty$. There are deterministic positive sequences ρ_n and $\gamma_n \rightarrow 0$ as $n \rightarrow \infty$ such that the minimum- $\mathcal{H}_{k_n}$ -norm interpolant is consistent for P in the sense of Eq. (9.2).*

The conclusion applies in fixed dimension. The deterministic tuning sequences are part of the procedure covered by the theorem. The mechanism follows from the Gram matrix of the narrow component:

$$\check{k}_{\gamma_n}(\mathbf{X}, \mathbf{X}) \approx I_n.$$

Write $\tilde{K} = (\tilde{k}(X_i, X_j))_{i,j}$. The smooth part of the interpolant then approximates

$$\tilde{k}(x, \mathbf{X})^\top (\tilde{K} + \rho_n I_n)^{-1} \mathbf{Y},$$

while the narrow part corrects the remaining training residuals. The displayed predictor is kernel ridge regression with penalty $\lambda_n = \rho_n/n$ when empirical squared loss is divided by n . Controlling the approximation and the population cost of the corrections completes the consistency argument. A nearly diagonal Gram matrix controls the sample-fitting part. This is the RKHS version of the point-corrected smoother and the weighted series rule. The two kernel theorems concern different procedures: [Theorem 9.4.2](#) uses a changing sum of a smooth kernel and a separate increasingly narrow kernel, whereas [Theorem 9.4.1](#) uses a single Laplace kernel with an adaptively chosen bandwidth.

9.5 What the results establish

What do these successes and failures establish about the original question, and what advice do they justify? First, a somewhat trivial observation is that interpolation is compatible with consistent regression when the selected solution contains two effective components:

1. a component whose bias and estimation error for the regression function vanish; and
2. enough localized or low-population-cost directions to fit the remaining noisy responses.

Second, the choice of features and the norm both matter. The scaled Legendre–Chebyshev construction is consistent under our assumptions, whereas the all-Legendre variant with the same scaling can be inconsistent (see [Endnote 5](#)). The spiky-smooth kernel supplies another positive example; [Theorem 9.4.1](#) gives another failure of minimum-norm interpolation, for the ordinary Laplace RKHS in fixed dimension.

These results take it for granted that interpolation needs to happen and show that under the right circumstances, interpolation is harmless. On the prescriptive side of things, there is no implication that interpolation would be desired. What may be desired though is “overparametrization” because adding modeling flexibility could in theory reduce approximation error, and such, the model could *adapt* to how challenging the learning problem is. However, as the results cited show, this is not automatic. For our Legendre–Chebyshev map, the features used to pick up signal are differently scaled from the rest and, as such, the effective capacity of the model remains the same as one gets from using only the Legendre basis block. To adapt to a target function’s unknown smoothness level, one would still need to use model-selection on the top of our procedure to adjust the number of Legendre basis functions. An oversized model capacity does not guarantee that that capacity will be used to model “signal” as opposed to capture noise. Moreover, we see that overfitting to the data may not even be competitive to the well-trodden path of using early stopping and/or regularization with model selection.

9.6 Does understanding deep learning require rethinking generalization? *

The interpolation results also help us assess the argument that motivated this lecture. What do the random-label experiments establish about existing generalization bounds, and what should we expect such bounds to explain?

9.6.1 The random-label argument

Based on their experiments, [Zhang et al. \[2017\]](#) draw the conclusion which gave the title of their paper: “Understanding deep learning requires rethinking generalization”. Their argument is that existing theoretical bounds on the generalization gap lack explanatory power in the experimental settings they consider. Here the gap is the difference between population and training classification error; the experiments estimate population error using a test set. This section concerns classification, in contrast to the squared-loss regression analysis above.

Their story goes as follows: in their experiments with pure noise labels, the gap is large, while with the original labels, the gap is small. To make the argument precise, consider a bound on the *expected* generalization gap for i.i.d. training observations and a fresh test observation from the same law, averaging over the sample and any independent algorithmic randomness. This theoretical comparison uses the same label law for training and testing; the paper instead evaluates corrupted-label training on the original test labels. Suppose the bound depends on sample size, the input distribution, and the learning algorithm, but is independent of the conditional distribution of labels given inputs. It must hold for both labeling distributions.

For example, suppose labels are independent of inputs and uniformly distributed over K classes. Every classifier has population error $1 - 1/K$. If the algorithm interpolates almost surely, its expected gap is therefore $1 - 1/K$, the maximal gap in this model. The bound must be at least this large. Now consider a labeling distribution under which the same algorithm has a small expected gap. The bound remains large, since it depends only on the input distribution and the algorithm. Hence it cannot certify the small gap in this second setting. This formalizes the limitation suggested by the experiments; a large upper bound does not predict that the actual gap must be large.

Quantities that take this form include the *VC dimension* of a fixed class of predictors and its Rademacher complexity. For binary classifiers, the VC dimension is the largest number of inputs on which the class can realize every binary labeling, or infinity if there is no finite maximum. Empirical Rademacher complexity measures the class’s ability to correlate with independent random signs on the observed inputs; its expectation depends on the input distribution, not the labeling distribution. One can associate these capacity measures with an algorithm by applying them to the fixed class of all predictors it can output. Uniform stability takes a different route: it controls an algorithm’s *worst-case* change in loss when one training observation is replaced, uniformly over training samples and test observations. Its distribution-free bound must cover random-label samples too. Thus a uniform-stability guarantee large enough to cover the large random-label gap cannot certify the small original-label gap. This is the comparison behind the paper’s stability objection, not a separate experiment measuring stability.

The conclusion concerns bounds with the specified independence from the labeling distribution. Averaging a data-dependent quantity does not automatically give it this property: for example, the expected norm or margin of a learned predictor can still depend on the labeling distribution.

9.6.2 What the argument does not rule out

Learning theorists had already developed bounds that use more information than these labeling-distribution-independent quantities. Examples include bounds evaluated using the learned weights' norms, the achieved margins, or a data-dependent distribution over predictors in a PAC-Bayes analysis. The labels can influence all these quantities. Fitting random labels does not establish that training on the original labels produces equally large norms or equally small normalized margins. The argument above therefore does not rule out useful bounds of these kinds. Some of them themselves use Rademacher complexity or uniform convergence, applied to suitably restricted classes. *Uniform convergence* controls the discrepancy between population and empirical losses simultaneously over a class of predictors. A limitation of the full architecture class is not a limitation of every use of these methods.

The paper acknowledges norm-based bounds, and Section 5 does compare norms in a linear/kernel experiment. Wavelet preprocessing increases the reported norm while decreasing test error. This demonstrates that the raw norm does not correctly order these two outcomes; it is not an evaluation of a norm-based generalization bound. Such bounds also account for the feature geometry and the loss or margin. In particular, changing the representation can change the norm assigned to a function: even rescaling all features changes the coefficient norm without changing predictions when coefficients are rescaled inversely. Wavelet preprocessing can change the kernel itself. A comparison must specify these choices. Moreover, even when two norm-based bounds are comparable, a larger upper bound need not accompany a larger actual risk. Inferring the failure of norm-based bounds from the reversed ordering alone is therefore invalid.⁹

It remains a substantive question whether any particular available bound gives a useful numerical guarantee in these experiments. The random-label argument and the raw-norm comparison do not settle it. Even if all available bounds had been labeling-distribution independent, this would justify seeking more informative guarantees for these experiments; it would not establish that generalization theory as a whole needs to be rethought.

9.6.3 What should we expect from a bound?

A bigger discussion topic, beyond this particular paper, is whether all generalization bounds need to be tight, or, more generally, just *how tight* they need to be. To answer this question, it is useful to consider the various goals theoreticians have in mind when developing their results.

The first goal, which I am often personally engaged with, is to gain understanding about just how difficult a certain *learning setting* is and what algorithms are most suitable for it. A learning setting specifies the learning instances, how algorithms gain information about them, and how performance is measured. VC theory, for example, characterizes the best achievable worst-case excess risk when competing with the best classifier in a specified class, under the chosen sampling and noise assumptions. Upper bounds are accompanied by lower bounds applying to every learner.

The second goal is to characterize the performance of specific algorithms or algorithm families. We can ask, for example, for an excess-risk bound for a support-vector machine or a kernel smoother. Margin-based, norm-based, and uniform-stability bounds can contribute to this goal.

The third goal is to obtain an informative guarantee, under prescribed assumptions, about the downstream performance of a learning method on particular data. These bounds are often data-dependent or empirical. A Hoeffding bound using an independent test set is the simplest example; more sophisticated examples include PAC-Bayes and leave-one-out bounds.

These three goals overlap. Bounds developed for the first goal inform us about the hardness of a learning setting. Bounds developed for the second inform us about the strengths and weaknesses of

an algorithm. For the third, the primary purpose is to certify performance. But a data-dependent bound can also contribute an explanation. For example, a margin bound can improve as the training margins increase even after training classification error has reached zero. This identifies a way that continued boosting can improve a guarantee without improving training classification error. It need not predict every change in the observed test error. The bibliographical notes give the classical source for this explanation.

Across all three goals, a useful answer requires a specification of the learning problems, the performance criterion, and the information available to the explanation. One algorithm may perform better on one distribution and worse on another, or the ordering may change with sample size. Comparing algorithms across problems therefore requires a criterion, such as worst-case risk over a specified class or average risk under a specified distribution of problems.

Generalization theory studies this dependence through selected properties of the data and learning procedure, such as sample size, norms, margins, or stability. Such abstractions can provide useful guarantees without preserving every distinction in actual risk. Different instances can agree on all the properties entering a bound yet have different risks. No quantity depending only on those properties can distinguish such instances. A bound can therefore be sharp in the worst case while substantially overestimating the risk in other cases. It is a misunderstanding of its role to require it to correctly rank every learning algorithm on every problem.

The restriction to selected properties is part of what makes the theory informative: a theorem identifies what those properties suffice to guarantee, without requiring a complete description of the learning problem. Allowing the bound to depend on the unknown risk itself would make an exact bound trivial, but explain nothing. Consequently, the inability of a particular bound to distinguish two experimental outcomes must be assessed against the claim that bound makes; it does not, by itself, establish the inadequacy of generalization theory.

9.6.4 What the interpolation results explain, and what to ask next

With this classification in place, we can approach the interpolation question in three ways. For the first goal, we can ask how well interpolating learners can do in the worst case over a specified class of instances. For the second, we can ask how well a particular interpolating learner does on different instances. For the third, we can seek data-dependent guarantees for the predictor an interpolating learner actually returns. Interpolation is one restriction on the algorithms that connects these questions to the experiments.

The results in this lecture contribute to the first two goals. They show that interpolation alone does not prevent consistency, and explain the performance of several specified selection rules. While these results do not apply to the trained neural networks in the experiments, they give a clear picture of what can govern linear interpolation in noisy regression. The learner needs an appropriate representation and norm: even linear transformations of the basis functions can matter. For the weighted polynomial rule, performance depends on tuning, how well the first feature block approximates the target, the number of features in that block, the amount of noise, and the population cost of fitting the residual with the second block.

The natural next question is how to carry out such an analysis for supplied features, rather than just the particular polynomials chosen here.

A question for further investigation. Fix an input law μ and a sequence of basis functions whose span is dense in $L^2(\mu)$. Under what additional conditions on the features and the coefficient penalties can we choose growing feature spaces so that the minimum-norm interpolants are consistent in probability for every $f^* \in L^2(\mu)$ and conditionally mean-zero noise of finite average variance? For

a specified smoothness class, when can the same construction, with suitable tuning, also attain the corresponding nonparametric rate? In the one-dimensional setting above, the benchmark for squared L^2 error is $n^{-2s/(2s+1)}$. Here universality concerns targets and noise laws with the fixed input law μ ; requiring one procedure to work for every input law is an additional question. Interpolation uses our convention for repeated inputs.

Density of the feature span only addresses approximation. The proof also needs reliable estimation on the first block, a well-conditioned sample Gram matrix for the residual block, and control of the population norm of its fitted residual. These requirements suggest which quantities a more general bound should contain. This is not a claim that the whole question is open: structured-feature and kernel results already give important sufficient conditions, as detailed in the bibliographical notes. The task for a new feature family is to determine whether those results apply and, where they do not, which additional assumptions or arguments are needed.

These observations motivate refinements and extensions of generalization theory, rather than establishing that the theory as a whole needs to be rethought. Classical least-squares theory already studied how prediction depends on the approximation space, sampling distribution, noise, and regularization. A general theory of when minimum-norm interpolation of noisy data generalizes was, as far as we know, undeveloped when the random-label paper appeared. The results discussed in this lecture address that specific question within the existing aims of learning theory.

9.7 Take-home messages

What should remain from the calculations when judging an interpolation claim?

- Nonemptiness of $\mathcal{A}(B)$ is too weak: every interpolating procedure succeeds on a deterministic one-point law. Seek nontrivial noisy members and nonmembers.
- A universally consistent regression procedure can be modified into an interpolating one with the same consistency class. Universal negative claims therefore require restrictions on the learning procedure or risk criterion.
- Exact fitting and small population squared error are compatible. The weighted polynomial rule shows how minimum-norm optimization can realize this compatibility in one fixed-feature model.
- The norm is part of the algorithm. In a series or RKHS, its coordinate weights determine whether the selected interpolant recovers the signal and where it places the fitted noise.
- Minimum norm can succeed or fail. Ordinary Laplace-kernel interpolation fails on a fixed smooth noisy problem; a deliberately structured spiky-smooth kernel is consistent.
- The weighted polynomial rule retains the classical $n^{-2s/(2s+1)}$ rate despite interpolation. Its raw guarantee is in probability; clipping yields expected-risk control when a valid target range is known.
- Advice to increase model size or training time requires comparison with regularized, smaller, and early-stopped procedures on the same problem.

Endnotes

1. Dimension in the illustration. The sample size is $n = 20$, the signal block has size $m = 4$, and the interpolation block has size $D = 8n = 160$. A generic feature space of total dimension at least n can already permit exact interpolation. The positive bound later contains an n/D term and additional conditioning requirements, so a larger interpolation block makes its population cost smaller. The displayed value makes the finite-sample contrast clear and is an illustrative choice. Smaller values of D also interpolate this sample, with a less pronounced separation. The asymptotic corollary uses the sufficient schedule $D_n = n^3$.
2. Scope of the illustration. A neural-network curve would specify one architecture, parameterization, optimizer, initialization, and stopping rule; a Laplace-kernel curve would specify one bandwidth. A single curve would not represent either family. The polynomial rules in the figure make the two selection procedures exact, and the main text analyzes their population behavior.
3. Classification and regression. The random-label experiments of [Zhang et al. \[2017\]](#) brought the same question into the study of deep networks. Let $\eta_k(x)$ be the clean conditional probability of class k . If a training label is replaced by a uniformly random class with probability p , then $\tilde{\eta}_k(x) = (1 - p)\eta_k(x) + p/K$. For $p < 1$, the most likely class is unchanged and every probability gap is multiplied by $1 - p$. An interpolating classifier fits the observed classes while the clean zero-one target remains the Bayes class. In squared-loss regression, the interpolator fits noisy responses while the population target remains the conditional mean. Zero-one loss asks for the conditional mode; squared loss asks for the conditional mean.
4. Point corrections. The functions $\mathbb{I}\{x = X_i\}$ are point indicators, sometimes informally called “spikes.”
5. An all-Legendre variant can fail consistency. Replace only the second-block functions χ_j by ψ_j , retaining the scaling in [Eq. \(9.10\)](#). For a fixed sample of distinct interior inputs and fixed m , as $D \rightarrow \infty$ the objective obtained by eliminating b converges to

$$\|\beta\|^2 + \frac{\pi}{2} \sum_{i=1}^n \sqrt{1 - X_i^2} (Y_i - u(X_i)^\top \beta)^2.$$

This is an unconstrained objective in β : the eliminated second block still supplies the correction needed to interpolate. Its residual weights are small near the endpoints. To see the consequence, define the weighted population projection

$$Q_m f = \operatorname{argmin}_{g \in \mathcal{F}_m} \int_{-1}^1 \sqrt{1 - x^2} |g(x) - f(x)|^2 \mu(dx).$$

With $m_n = \lfloor n^{1/4} \rfloor$, $D_n = n^3$, and noiseless responses $Y_i = f^*(X_i)$ for bounded f^* , the all-Legendre interpolator $\hat{f}_n^{\text{Leg}}$ satisfies

$$\|\hat{f}_n^{\text{Leg}} - Q_{m_n} f^*\|_{L^2(\mu)} \xrightarrow{\mathbb{P}} 0.$$

This comparison uses Legendre kernel estimates and concentration of the weighted design; the fixed-sample limit alone would not justify it. However, $Q_m f^*$ need not converge in the unweighted norm. An explicit bounded continuous example is

$$N_k = 2^{k^6}, \quad f^*(x) = \sum_{k=1}^{\infty} \frac{\sin(N_k \arccos x)}{k^2}.$$

The series converges uniformly, and $f^*(-1) = f^*(1) = 0$. Indeed, the polynomials $q_r(\cos \theta) = \sin(r\theta)/\sin \theta$ are orthogonal for the weighted projection, while $\|q_r\|_{L^2(\mu)}^2 = \sum_{j=1}^r (2j-1)^{-1} \sim \frac{1}{2} \log r$. The coefficient of q_{N_k} in this example is asymptotic to $2/(\pi k^2)$, so

$$\|Q_{N_k} f^* - Q_{N_k-1} f^*\|_{L^2(\mu)} \asymp k.$$

At least one of the two projection errors therefore diverges along a subsequence. Every integer m occurs as m_n (take $n = m^4$), so the finite-sample comparison gives failure of consistency for the interpolator. The difficulty concerns neighborhoods of the endpoints; changing values at the two endpoints cannot change an $L^2(\mu)$ error.

6. A bounded orthonormal polynomial example. If the fixed input law is the arcsine law and both feature groups use χ_j , then the proof gives $\mathbb{P}(\mathcal{G}^c) \leq Cn^2/D + m \exp(-cn/m)$. The same rate follows for every $s > 0$ satisfying the corresponding $L^2(\nu)$ approximation condition. This changes the input law; it does not remove the restriction in the uniform-input theorem.
7. Why three dimensions, and why Laplace? We specialize the original theorem to $d = 3$ for a concrete example; it covers every fixed odd dimension. The Laplace kernel is convenient for its proof because, on $\mathbb{R}^d$ with d odd, its squared RKHS norm is a weighted sum of integrals of squared derivatives through order $(d+1)/2$, including the function itself. For $d = 3$ this means derivatives through order two. This expression lets the proof turn norm control into control of variation between observations.
8. A conjectured approximation–noise trade-off. Consider the all-Legendre variant in [Endnote 5](#), with deterministic positive integer sequences m_n, D_n satisfying $m_n + D_n \geq n$ and the prescribed second-block scaling $D_n^{-1/2}$. The conjecture is: if the resulting interpolator is $L^2(\mu)$ -consistent in probability for every continuous target $f : [-1, 1] \rightarrow \mathbb{R}$ with noiseless observations $Y_i = f(X_i)$, then it is inconsistent for pure noise $Y_i = \varepsilon_i$, where the ε_i are independent $N(0, \sigma^2)$ variables, independent of the inputs, and $\sigma^2 > 0$ is fixed. Here μ is uniform on $[-1, 1]$, as in the polynomial example. The same sequences are used in both experiments.

Conditional on $X_1, \dots, X_n$, write the interpolator as $T_n \mathbf{Y}$, where $T_n : \mathbb{R}^n \rightarrow L^2(\mu)$ is linear. If $e_1, \dots, e_n$ are the standard coordinate vectors, its conditional squared error on pure noise is

$$\mathbb{E}[\|T_n \boldsymbol{\varepsilon}\|_{L^2(\mu)}^2 \mid X_1, \dots, X_n] = \sigma^2 \sum_{i=1}^n \|T_n e_i\|_{L^2(\mu)}^2.$$

For Gaussian noise, consistency in probability is equivalent to the sum on the right tending to zero in probability. The conjecture therefore asserts that universal noiseless consistency prevents this noise contribution from vanishing. polynomial can hold both with and without noise. Neither the preceding counterexample nor [Theorem 9.4.1](#) establishes this conjecture.

9. Norm bounds for linear and kernel classifiers. Such bounds were available before the random-label experiments. To see what enters them, fix a bounded feature map ϕ into a Hilbert space, with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$, and set $k(x, x') = \langle \phi(x), \phi(x') \rangle$. For binary labels $Y_i \in \{-1, 1\}$ and scores $f_w(x) = \langle w, \phi(x) \rangle$, a margin bound for $\|w\| \leq M$, with fixed $M > 0$ and margin threshold $\gamma > 0$, combines empirical margin loss, a complexity term of order

$$\frac{M}{\gamma n} \left(\sum_{i=1}^n k(X_i, X_i) \right)^{1/2},$$

and a confidence term. Empirical margin loss here means the sample average of the ramp loss $\min\{1, \max\{0, 1 - Y_i f_w(X_i)/\gamma\}\}$. Replacing ϕ by $a\phi$ and w by w/a , for $a > 0$, leaves the margins and this complexity term unchanged when M is rescaled by $1/a$. The coefficient norm alone is not invariant. Bounds of this form follow from Rademacher complexity; a classical source is identified in the bibliographical notes.

Bibliographical notes

The motivating experiments are due to [Zhang et al. \[2017\]](#), first circulated in 2016. They independently replace training labels by uniformly random classes with a chosen probability, while evaluating on the original test labels. Complete corruption demonstrates memorization capacity. Partial corruption demonstrates simultaneous fitting of injected noise and useful prediction from the remaining signal. Symmetric corruption below complete corruption preserves the Bayes class and shrinks class-probability gaps, so the gradual change in test error is compatible with the population problem. The paper helped bring noisy interpolation to the attention of learning theorists. Its Sections 1.2 and 2.2 discuss capacity and stability bounds, and Section 5 compares norms of linear/kernel interpolants. The optional discussion in [Section 9.6](#) examines the conclusions drawn from these comparisons.

Classical learning theory beyond architecture-level capacity. The book by [Anthony and Bartlett \[1999\]](#) is a useful starting point. Its treatment of margins, scale-sensitive capacity, parameter norms, and model selection distinguishes nominal network size from the quantities that enter a generalization guarantee. It is therefore incorrect to identify classical learning theory with coarse uniform-convergence bounds over the full class represented by an architecture. Some finer bounds also use uniform convergence. In particular, [Bartlett and Mendelson \[2002\]](#) develop Rademacher-based bounds for several function classes; their Theorem 21 gives the kernel margin bound discussed in [Endnote 9](#). These are bounds on classification performance, not assumption-free guarantees for unbounded squared-loss regression. [Zhang et al. \[2017\]](#) themselves acknowledge earlier norm-based capacity bounds in Sections 1.2 and 2.2, so the issue is what their experiments establish about those bounds, not whether such bounds were known to them.

The margin analysis of [Schapire et al. \[1998\]](#) is a classical example of a data-dependent bound contributing an explanation. Its dependence on the empirical margin distribution explains how a voting classifier’s guarantee can improve after training classification error is zero. It does not assert that test error must decrease at every subsequent boosting round. [Wyner et al. \[2017\]](#), first circulated in 2015, later described AdaBoost and random forests as interpolating classifiers and proposed a spiked-smooth mechanism. These classification results already connected interpolation with good prediction, but did not settle minimum-norm interpolation of noisy regression responses.

From nonparametric prediction to minimum-norm interpolation. Nonparametric regression had long analyzed procedures whose approximation spaces grow with sample size. [Györfi et al. \[2002\]](#) develop consistency and rate results for nearest-neighbor, kernel, and series estimators, including the role of smoothing and approximation. Classical least-squares analyses also account for the representation and input law: [Cohen et al. \[2013\]](#) study general approximation spaces under random sampling, while [Hsu et al. \[2012\]](#) give random-design prediction bounds for ordinary least squares and ridge regression involving covariance estimation, approximation error, and noise. These results supply a starting point for asking the corresponding questions about noisy minimum-norm interpolation.

For high-dimensional linear regression, [Bartlett et al. \[2020\]](#) used the term *benign overfitting* and characterized spectral conditions under which the minimum-Euclidean-norm interpolator has small excess risk despite response noise. Their random-design assumptions and covariance conditions are not automatically satisfied by a fixed polynomial feature map. [Liang and Rakhlin \[2020\]](#) and [Liang et al. \[2020\]](#) give positive risk bounds for minimum-norm kernel interpolation when input dimension grows with sample size, a different asymptotic regime from this lecture’s fixed-dimensional polynomial example. [Belkin \[2021\]](#) surveys the interpolation literature and its relation to older statistical ideas.

The lecture’s results and their sources. The lecture separates the existence of consistent interpolators from the analysis of a specified selection rule. The elementary point-correction argument in [Proposition 9.2.1](#), combined with a universally consistent base estimator, gives an interpolating procedure whose benign set is the base estimator’s full consistency class. Response averaging handles point masses under our interpolation convention. The base estimator can be chosen from the classical procedures treated by [Györfi et al. \[2002\]](#). More substantive interpolating nonparametric procedures appear in [Belkin et al. \[2019\]](#), who construct singular-kernel estimators that interpolate and attain standard nonparametric rates. These are positive existence results for deliberately constructed estimators; they do not analyze the predictors selected by ordinary neural-network training.

For a specified minimum-norm rule, [Theorem 9.3.1](#) identifies how one chosen feature scaling permits both signal recovery and interpolation of noise, while [Proposition 9.3.4](#) shows failure with a different norm on the same polynomial space. These are stylized explanations for explicit feature choices, not a characterization of prediction risk for arbitrary feature families. Both results were derived for these notes; they are not restatements of the published results cited here. Priority for these particular formulations has not been established. Related published minimum-Hilbert-norm results are given by [McRae et al. \[2022\]](#). Their structured-feature framework includes bounded orthonormal systems such as Fourier and Chebyshev features and provides nonasymptotic conditions for small regression risk. Theorems 1 and 2 bound bias and noise contributions using conditioning of the signal-block sampling operator and the residual Gram matrix; Section 2.3 gives conditions on features that ensure this conditioning. Their stated bias theorem assumes the target lies in the chosen finite-dimensional signal space. Thus it is a concrete starting point for the supplied-feature question in [Section 9.6](#), but does not by itself give the all- L^2 consistency statement asked for there. Extending the target class requires controlling the approximation residual, as in this lecture’s proof. The spiky-smooth construction below supplies another positive answer for a deliberately chosen kernel family, not for arbitrary supplied features.

The polynomial proof combines the estimates in [Lemma 9.B.1](#), matrix Chernoff concentration in the form developed by [Tropp \[2012\]](#), and classical approximation theory. [DeVore and Lorentz \[1993\]](#) treat Jackson estimates in uniform and L^p norms. For integer $s \geq 1$, the L^2 estimate gives approximation error of order m^{-s} for functions in $W^{s,2}([-1,1])$, hence squared error of order m^{-2s} ; the uniform Jackson estimate gives the Lipschitz example. The stability analysis of [Cohen et al. \[2013\]](#) complements these approximation estimates: in their uniform-input Legendre example, the sum of squared basis functions has supremum m^2 , which explains the quadratic sample-size condition used here. [Györfi et al. \[2002, Chapters 11 and 18\]](#) treat least-squares rates and orthogonal-series estimation. The exponent in [Corollary 9.3.3](#) is the classical nonparametric exponent: [Stone \[1982\]](#) establishes $2s/(2s+1)$ for squared L^2 error in one dimension. This minimax benchmark concerns the worst case over a bounded smoothness class, with a regular design law and nondegenerate noise. Choosing m_n using s is not adaptation to unknown smoothness. These claims distinguish

approximation, statistical estimation, and stability of a particular fitting method.

Two refinements clarify the scope of the polynomial results. First, the projection obstruction in [Endnote 5](#) explains why the choice of the second feature block matters. The obstruction is classical: [Muckenhoupt \[1969, Theorem 2\]](#) gives the failure of unweighted L^2 mean convergence for Jacobi expansions with parameters $\alpha = \beta = 1/2$. The fixed-sample kernel limit and its finite-sample control use the uniform Legendre asymptotics recorded in [NIST \[2026, §14.15\(iii\)\]](#), together with matrix concentration. The explicit continuous target and its application to the scaled all-Legendre interpolator were derived for these notes. No earlier source for this particular interpolation counterexample has been identified; its novelty has not been established. The tuning conjecture in [Endnote 8](#) is posed here as an unresolved question, not as a consequence of the cited inconsistency results.

Second, convergence in probability does not by itself give convergence of expected risk. [Proposition 9.C.1](#) gives a direct close-pair argument in a general function class with norm-controlled variation. It is not claimed as a new result. Truncation of least-squares estimates and the distinction between convergence in probability and expected risk are classical topics; [Györfi et al. \[2002, Chapters 10–11 and 18\]](#) is a reference for truncated least squares and orthogonal-series estimation.

The kernel results give further success and failure comparisons for specified minimum-norm procedures. For standard kernel families and their smoothness properties, including the Matérn family, see [Rasmussen and Williams \[2006, Chapter 4\]](#); Chapter 5 discusses kernel selection and hyperparameter tuning, including cross-validation. The preceding guidance on kernel choice concerns approximation and regularized prediction, not a general guarantee for noisy interpolation. The universality of Laplace and Gaussian kernels, used here only as an approximation property on compact domains, follows from the criteria in [Sriperumbudur et al. \[2010\]](#). The fixed-problem negative result in [Theorem 9.4.1](#) specializes [Rakhlin and Zhai \[2019, Theorem 1\]](#). Their theorem allows fixed odd dimension, an input density on the unit ball bounded above and below, a fixed nonzero smooth regression function, and independent uniform-sign noise. The constant target used here has a smooth compactly supported extension, as required by their proof. [Buchholz \[2022\]](#) extends fixed-dimensional inconsistency to additional Sobolev kernels, and [Beaglehole et al. \[2023\]](#) give expected-risk inconsistency results for translation-invariant kernels, including periodic versions of Gaussian kernels, on a fixed grid with noisy responses. This differs from the i.i.d. sampling and in-probability lower bound in [Theorem 9.4.1](#); the latter theorem should not be read as a statement about every standard kernel. [Haas et al. \[2023\]](#) prove both broader fixed-dimensional obstructions for smooth estimators and positive spiky-smooth results. [Theorem 9.4.2](#) records the existence conclusion of their Theorem 8 for a Laplace correction, without reproducing a tuning schedule. When translating a schedule, distinguish an additive Gram-matrix shift ρ_n from the normalized empirical-loss penalty ρ_n/n . They also translate their kernel construction into special neural-network kernels and activation functions. This supplies a neural connection, but the mechanism is still an explicitly designed signal-plus-spike geometry.

Finally, the negative result of [Kohler and Krzyżak \[2021\]](#) has a different scope: it concerns estimators that nearly minimize empirical squared loss and a worst-case choice of law. For every sufficiently large n , their lower bound supplies a law in a smooth-regression class on which the expected L^2 error is bounded below by a constant. The law has a finite input support and may change with n . Consequently, the result rules out a uniform risk guarantee over that class; it does not exhibit one fixed law P on which the estimator is inconsistent, and therefore does not show that $P \notin \mathcal{A}(B)$ for any fixed P . In contrast, [Theorem 9.4.1](#) gives a fixed- P inconsistency statement for Laplace minimum-norm interpolation.

Glossary

Which terms distinguish the procedure, the fitting requirement, and the risk guarantee?

Regression function f_P^*

The conditional mean $f_P^*(x) = \mathbb{E}[Y \mid X = x]$, defined up to P_X -null sets. It minimizes population squared-loss risk.

Population norm $\|f\|_{L^2(\mu)}$

The norm $\|f\|_{L^2(\mu)} = (\int f^2 d\mu)^{1/2}$ under the input law μ , also abbreviated $\|f\|_2$ here. The space $L^2(\mu)$ identifies square-integrable functions that agree μ -almost everywhere; changing values on a set of input probability zero does not change this norm.

Population excess risk

Population risk minus the smallest achievable population risk. Under squared loss it equals $\|f - f_P^*\|_{L^2(P_X)}^2$ by [Eq. \(9.1\)](#). For a learned predictor, it is random through the training sample.

Consistency in probability and in expected risk

For a fixed law P , consistency in probability means $\|\hat{f}_n - f_P^*\|_{L^2(P_X)}^2 \xrightarrow{\mathbb{P}} 0$. Expected-risk consistency requires $\mathbb{E}[\|\hat{f}_n - f_P^*\|_{L^2(P_X)}^2] \rightarrow 0$. The latter implies the former, but the converse need not hold.

Universal consistency

Consistency of one fixed learning procedure for every law in the stated problem class. The procedure does not depend on the unknown law, and convergence need not be uniform over the class.

Benign set $\mathcal{A}(B)$

The fixed problem laws on which the specified interpolating procedure B has population excess risk converging to zero in probability. The expected-risk version is denoted by $\mathcal{A}_E(B)$ in [Appendix 9.C](#).

Benign overfitting

Exact fitting of the observed responses together with vanishing population excess risk, including when the responses contain nondegenerate noise.

Interpolation

Exact satisfaction of the empirical fitting constraints; here $f(X_i) = Y_i$ at every nonconflicting training input. An index is nonconflicting when all observations at its input have the same response.

Minimum-norm interpolant

The feasible interpolating predictor having smallest norm in a specified feature or function space. The norm and representation are part of the procedure.

Feature-induced function norm $\|f\|_\psi$

The smallest Euclidean coefficient norm among representations $f(x) = \theta^\top \psi(x)$. Changing the features or their scaling can change this norm and the selected interpolant without changing the represented function space.

Non-atomic input law

An input law with no atom: no measurable set A of positive probability has the property that every measurable subset of A has probability either zero or $P_X(A)$. For Borel laws on Euclidean spaces, this is equivalent to $P_X(\{x\}) = 0$ for every x . An i.i.d. sample from such a law has distinct inputs almost surely.

Approximation error $A_m(f^*)$

The smallest squared $L^2(\mu)$ error achievable by a polynomial of degree below m : $A_m(f^*) = \inf_{g \in \mathcal{F}_m} \|g - f^*\|_{L^2(\mu)}^2$.

Polynomial approximation order s

The condition $A_m(f^*) \leq L^2 m^{-2s}$ for every $m \geq 1$, with some fixed $0 \leq L < \infty$. The corresponding best norm error is at most Lm^{-s} .

Minimax-optimal rate

A rate achieved uniformly over a specified problem class and matched, up to constants, by a lower bound for every learning procedure, under the specified risk criterion. No estimator achieves a faster rate uniformly over that class; the difficult problem may change with n .

Orthonormal basis

An orthogonal family of unit-norm functions whose finite linear combinations are dense in the space under discussion. Orthogonality and norms are relative to the specified inner product.

Orthogonal projection

The unique closest element of a closed linear subspace in the Hilbert-space norm. The approximation residual is orthogonal to the subspace. Here, g_m is the projection of f^* onto $\mathcal{F}_m$ in $L^2(\mu)$.

Positive semidefinite kernel

A symmetric function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ whose matrix of values on any finite set of inputs is positive semidefinite. In a feature representation, $k(x, x') = \langle \varphi(x), \varphi(x') \rangle$.

Reproducing kernel Hilbert space (RKHS)

A Hilbert space $\mathcal{H}_k$ of functions with continuous point evaluations, represented by a kernel through $f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}_k}$. This identity is the reproducing property. Its norm determines which interpolant the minimum-norm rule selects.

Representer theorem

In the interpolation form used here, whenever the constraints are feasible, the minimum-RKHS-norm interpolant lies in $\text{span}\{k(X_1, \cdot), \dots, k(X_n, \cdot)\}$. Thus the possibly infinite-dimensional optimization reduces to a finite kernel-matrix problem; see Eqs. (9.29) and (9.30).

Kernel Gram matrix

The matrix $K = (k(X_i, X_j))_{i,j=1}^n$ of kernel values at the sample inputs. Its entries are inner products of the corresponding feature vectors.

Kernel integral operator

The map $T_{k,\mu}$ in Eq. (9.25), which depends on the kernel and input law. In the orthonormal expansion Eq. (9.26), the RKHS norm weights squared coefficients by the reciprocals of the positive eigenvalues, whereas the $L^2(\mu)$ norm weights them equally.

Universal kernel

On a compact domain $\mathcal{X}$, a continuous kernel whose RKHS is dense in the continuous functions on $\mathcal{X}$ under the uniform norm. This is an approximation property of a kernel; universal consistency is a property of a learning procedure.

Linear and polynomial kernels

The kernels $k(x, x') = x^\top x'$ and $k(x, x') = (1 + x^\top x')^q$, respectively, with positive integer q . The former uses the original coordinates as features; the latter represents monomials through degree q with specific weights. Both have finite-dimensional feature spaces.

Radial kernel

A kernel whose value depends only on the Euclidean distance $\|x - x'\|$. Laplace and Gaussian kernels are examples.

Bandwidth (length scale)

The distance-scale parameter $\gamma > 0$ in kernels such as $\exp(-\|x - x'\|/\gamma)$ and $\exp(-\|x - x'\|^2/(2\gamma^2))$. Smaller bandwidths give more localized kernels; larger bandwidths give slower decay with distance. In the parametrization $k_c(x, x') = c^3 \exp(-c\|x - x'\|)$, the bandwidth is $1/c$, not c .

Laplace kernel

The radial kernel $k(x, x') = \exp(-\|x - x'\|/\gamma)$, with bandwidth $\gamma > 0$. It has an infinite-dimensional RKHS and is universal on compact domains, but its minimum-norm interpolant need not be consistent.

Gaussian kernel

The radial kernel $k(x, x') = \exp(-\|x - x'\|^2/(2\gamma^2))$, also called the squared-exponential kernel, with bandwidth $\gamma > 0$. It has an infinite-dimensional RKHS and is universal on compact domains.

Matérn family

A family of radial kernels with a smoothness parameter $\nu > 0$ and a separate bandwidth $\gamma > 0$. It includes the Laplace kernel at $\nu = 1/2$ and approaches the Gaussian kernel as $\nu \rightarrow \infty$ with γ fixed. Increasing ν increases smoothness; changing γ changes the distance scale.

Kernel ridge regression

Squared-loss regression with a positive penalty $\lambda \|f\|_{\mathcal{H}_k}^2$ added to the average training loss. With this normalization, its kernel-matrix formula uses $K + n\lambda I_n$ in place of K . Positive regularization permits nonzero training residuals.

Ridgeless kernel regression

Minimum-RKHS-norm interpolation, equivalently the zero-regularization limit of kernel ridge regression when the kernel matrix is invertible.

Spiky-smooth kernel

A kernel formed from a smooth component for learning the regression function and a sharply localized component for satisfying interpolation constraints.

Clipping

Replacing a prediction $f(x)$ by $\max\{-M, \min\{M, f(x)\}\}$. If $|f^*| \leq M$, this cannot increase squared prediction error. It preserves interpolation when the observed responses lie in $[-M, M]$.

Generalization gap

The population risk of a learned predictor minus its average training loss. In the optional discussion these are classification error rates. This differs from excess risk, which subtracts the optimal population risk rather than training loss.

VC dimension

For a class of binary classifiers, the largest number of inputs on which the class can realize every binary labeling, or infinity if unbounded. It is a property of the class, not of the predictor selected from it.

Empirical Rademacher complexity

The expected supremum of the average correlation between functions in a specified class and independent uniform random signs on the observed inputs. The expectation here is over the signs; averaging also over the inputs gives the corresponding population Rademacher complexity. For a fixed class, neither quantity uses the labeling distribution.

Uniform convergence

Control of the difference between population and empirical losses simultaneously for every predictor in a specified class. This allows the bound to apply to a predictor selected from that class using the training data.

Uniform stability

A uniform bound on the change in an algorithm’s test loss when one training observation is replaced. The bound must hold over all training samples and test observations, not just the observed sample.

Rate in probability $O_{\mathbb{P}}(a_n)$

For deterministic $a_n > 0$, $Z_n = O_{\mathbb{P}}(a_n)$ means that, for every $\delta > 0$, some constant C_δ satisfies $\mathbb{P}(|Z_n| > C_\delta a_n) \leq \delta$ for all sufficiently large n . This does not assert an expected-value bound.

9.8 Exercises

Can you derive the squared-loss identity and then use the test measure and the chosen norm to predict when interpolation preserves a signal? The exercises also ask you to justify the projection, optimization, and matrix calculations in the main proof, and to extend consistency to finite average noise variance.

General instructions. Justify every answer, including yes/no answers, even when the question does not explicitly ask for a justification.

Exercise profiles. Filled mountains show difficulty on a four-mountain scale: one is a direct application, two uses a familiar method, three combines ideas, and four is a substantial challenge. Filled cogs show technical load on a three-cog scale: one is light, two is sustained, and three is dense. Filled hourglasses show estimated time: one is under 20 minutes, two is 20–45 minutes, three is 45–90 minutes, and four is more than 90 minutes. Pale pictograms show the unused part of each scale. The work row uses the canonical order  proof,  calculation,  construction,  computation, and  interpretation; pale icons mark work types not used.  marks the intended learning outcome. Time estimates cover the non-optional parts. Ratings are not prerequisites or grading weights. A superscript asterisk marks an exercise based on supplementary lecture material; it is not a difficulty rating and does not make the exercise optional when assigned.

Exercise 1: The squared-loss excess-risk identity.

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 ☉ Derive the population identity that turns squared-loss excess risk into L^2 estimation error.

Let $P \in \mathcal{P}_2$, write $\mu = P_X$, and let $f_P^*(X) = \mathbb{E}[Y | X]$. For any fixed $f \in L^2(\mu)$, prove

$$\mathbb{E}[(f(X) - Y)^2] - \mathbb{E}[(f_P^*(X) - Y)^2] = \|f - f_P^*\|_{L^2(\mu)}^2.$$

Then let S_n be a training sample independent of the test pair (X, Y) and let $\hat{f}_n$ be selected from S_n . Show that, whenever the displayed quantities are finite, the same identity holds conditional on S_n with $f = \hat{f}_n$.

Exercise 2: Point corrections and the test measure.

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 ☉ Separate exact fitting at the observations from integrated prediction error.

Let μ be a Borel probability law on $\mathcal{X} \subset \mathbb{R}$, let $x_1, \dots, x_n \in \mathcal{X}$ be distinct, and let g, f^* be measurable, square-integrable functions with specified point values. For numbers $r_1, \dots, r_n$, define

$$h(x) = g(x) + \sum_{i=1}^n r_i \mathbb{I}\{x = x_i\}.$$

(a) Suppose that $\mu(\{x\}) = 0$ for every $x \in \mathcal{X}$. Prove

$$\|h - g\|_{L^2(\mu)} = 0, \quad \|h - f^*\|_{L^2(\mu)} = \|g - f^*\|_{L^2(\mu)}.$$

Choose r_i so that $h(x_i) = y_i$ for arbitrary prescribed values $y_1, \dots, y_n$.

(b) Show that the conclusion can fail when the test law has a point mass. Give an explicit choice of $\mathcal{X}, \mu, g, f^*, x_1, r_1$ for which

$$\|h - f^*\|_{L^2(\mu)}^2 > \|g - f^*\|_{L^2(\mu)}^2.$$

Show that the distinction from part (a) is whether the set on which the correction is nonzero has zero or positive μ -measure, and state why this determines whether the correction contributes to the population $L^2(\mu)$ error.

(c) **Optional.** Let $P \in \mathcal{P}_2$ be arbitrary, and assume that every singleton in $\mathcal{X}$ is measurable. Let $S_n = ((X_i, Y_i))_{i=1}^n$ be an i.i.d. sample with $\mathcal{L}(X_i, Y_i) = P$. Write $N_n(x) = \sum_{i=1}^n \mathbb{I}\{X_i = x\}$ and define

$$\hat{f}_n(x) = \begin{cases} N_n(x)^{-1} \sum_{i: X_i=x} Y_i, & N_n(x) > 0, \\ \tilde{f}_n(x), & N_n(x) = 0, \end{cases}$$

where $\tilde{f}_n$ is any estimator. Prove that, for every training index i such that $X_j = X_i$ implies $Y_j = Y_i$ for all $j \in [n]$,

$$\hat{f}_n(X_i) = Y_i.$$

(d) **Optional.** In the setting of part (c), suppose that

$$\|\tilde{f}_n - f_P^*\|_{L^2(P_X)}^2 \xrightarrow{\mathbb{P}} 0.$$

Prove

$$\|\hat{f}_n - f_P^*\|_{L^2(P_X)}^2 \xrightarrow{\mathbb{P}} 0 \quad \text{as } n \rightarrow \infty.$$

You may use that the set $\mathcal{D} = \{x : P_X(\{x\}) > 0\}$ is countable, the strong law of large numbers, and dominated convergence for sums over $\mathcal{D}$.

HINTS For (b), a one-point space is sufficient.

Exercise 3: Minimum-norm solutions and elimination of a coefficient block.

⊙ Derive the minimum-norm formula and reduce a constrained fit to a quadratic problem.

All vector norms below are Euclidean.

- (a) Let $M \in \mathbb{R}^{n \times p}$ and suppose $M\theta = y$ is feasible. Prove that the feasible vector of minimum norm exists and is unique. When M has row rank n , prove that

$$\hat{\theta} = M^\top (MM^\top)^{-1}y, \quad \|\hat{\theta}\|^2 = y^\top (MM^\top)^{-1}y.$$

Show that every feasible θ has the form $\hat{\theta} + z$ with $z \in \ker M$, and that $\|\hat{\theta} + z\|^2 = \|\hat{\theta}\|^2 + \|z\|^2$.

- (b) Let $U \in \mathbb{R}^{n \times m}$, let $V \in \mathbb{R}^{n \times D}$ have row rank n , and set $H = D^{-1}VV^\top$. For fixed $\beta \in \mathbb{R}^m$, derive

$$b_\beta = V^\top (VV^\top)^{-1}(\mathbf{Y} - U\beta), \quad D\|b_\beta\|^2 = (\mathbf{Y} - U\beta)^\top H^{-1}(\mathbf{Y} - U\beta).$$

Deduce that minimizing $\|\beta\|^2 + D\|b\|^2$ subject to $U\beta + Vb = \mathbf{Y}$ reduces to minimizing

$$q(\beta) = \|\beta\|^2 + (\mathbf{Y} - U\beta)^\top H^{-1}(\mathbf{Y} - U\beta).$$

- (c) Differentiate q to prove

$$\hat{\beta} = (I_m + U^\top H^{-1}U)^{-1}U^\top H^{-1}\mathbf{Y}.$$

Explain why the inverse exists even when U has dependent columns. Writing $A = U^\top H^{-1}U$ and $T = (I_m + A)^{-1}U^\top H^{-1}$, verify

$$TU = I_m - (I_m + A)^{-1},$$

and derive Eq. (9.20) from Eq. (9.19).

HINTS For (a), decompose a feasible vector into its components in $\ker M$ and $(\ker M)^\perp$. For (c), examine the quadratic form of $I_m + A$.

Exercise 4: The function norm induced by a feature map.

⊙ Relate parameter minimization to a function norm even when representations are not unique.

Let $\psi : \mathcal{X} \rightarrow \mathbb{R}^p$, let $\mathcal{F} = \{x \mapsto \theta^\top \psi(x) : \theta \in \mathbb{R}^p\}$, and define

$$N = \{\theta : \theta^\top \psi(x) = 0 \text{ for every } x \in \mathcal{X}\}, \quad \|f\|_\psi = \inf_{\theta: \theta^\top \psi = f} \|\theta\|.$$

Equality here is pointwise on $\mathcal{X}$.

- (a) Prove that every $f \in \mathcal{F}$ has a unique representing vector $\theta_f \in N^\perp$, that all its representing vectors form $\theta_f + N$, and that $\|f\|_\psi = \|\theta_f\|$. Show that $f \mapsto \theta_f$ is linear and deduce that $\|\cdot\|_\psi$ is a norm.
- (b) Suppose the interpolation constraints $f(X_i) = Y_i$ are feasible. Prove that minimizing $\|\theta\|$ over interpolating parameters and minimizing $\|f\|_\psi$ over interpolating functions give the same unique function. Establish existence and uniqueness, including when the features are linearly dependent.

- (c) For invertible $G \in \mathbb{R}^{p \times p}$, set $\psi^{(G)} = G\psi$. Prove that the represented function space is unchanged and that

$$\|f\|_{\psi^{(G)}} = \inf_{\theta: \theta^\top \psi = f} \|G^{-\top} \theta\|.$$

Explain why, when the features are dependent, this infimum need not be attained at the vector θ_f from (a).

HINTS Use the orthogonal decomposition $\mathbb{R}^p = N \oplus N^\perp$. For (c), the coordinate change is $\theta = G^\top c$.

Exercise 5: Best approximation, orthogonality, and synthesis bounds.

- 🕒 Derive the population-norm identities used to separate approximation and estimation error.

Use the inner product and norm from [Appendix 9.A](#). Let $\psi_0, \dots, \psi_{m-1}$ be orthonormal in $L^2(\mu)$, $\mathcal{F}_m = \text{span}\{\psi_0, \dots, \psi_{m-1}\}$, and $f^\star \in L^2(\mu)$.

- (a) Prove that the unique best approximation $g_m \in \mathcal{F}_m$ is

$$g_m = \sum_{j < m} \langle f^\star, \psi_j \rangle_{L^2(\mu)} \psi_j.$$

Writing $r = f^\star - g_m$, prove that $\langle r, g \rangle_{L^2(\mu)} = 0$ for every $g \in \mathcal{F}_m$. Also show directly that optimality forces this orthogonality by considering $\|f^\star - (g_m + tg)\|_{L^2(\mu)}^2$ for real t .

- (b) If $g_m = \beta_\star^\top u$ with $u = (\psi_0, \dots, \psi_{m-1})^\top$, prove

$$\|\beta_\star\|^2 = \|g_m\|_{L^2(\mu)}^2, \quad \|f^\star\|_{L^2(\mu)}^2 = \|g_m\|_{L^2(\mu)}^2 + \|r\|_{L^2(\mu)}^2.$$

For arbitrary $\hat{\beta} \in \mathbb{R}^m$ and $h \in L^2(\mu)$, deduce

$$\|\hat{\beta}^\top u + h - f^\star\|_{L^2(\mu)}^2 \leq 3\|\hat{\beta} - \beta_\star\|^2 + 3\|h\|_{L^2(\mu)}^2 + 3\|r\|_{L^2(\mu)}^2.$$

- (c) Suppose χ_j are orthonormal under a probability measure ν and $\mu(E) \leq M\nu(E)$ for every measurable set E , with $M < \infty$. Prove, for every finite coefficient vector b , that

$$\left\| \sum_j b_j \chi_j \right\|_{L^2(\mu)}^2 \leq M \sum_j b_j^2.$$

Specialize to the uniform and arcsine laws in the lecture to obtain [Eq. \(9.9\)](#).

HINTS Expand the squared norms using the inner product. For (c), extend the measure inequality from indicators to nonnegative simple functions and then to nonnegative measurable functions.

Exercise 6: Feature scaling can change the minimum-norm interpolant.

- 🕒 Construct a two-parameter example in which feature scaling changes the minimum-norm interpolant.

Using two parameters, construct two feature maps that represent the same function class but whose minimum-Euclidean-parameter-norm interpolants are different under the same interpolation constraint.

Exercise 7: A large penalty and two-stage fitting.

Quantify when a weighted interpolator approximates fitting the first block and then its residual.

Fix $U \in \mathbb{R}^{n \times m}$ of column rank m , $V \in \mathbb{R}^{n \times D}$ of row rank n , and $\mathbf{Y} \in \mathbb{R}^n$. For $\lambda > 0$, let $(\hat{\beta}_\lambda, \hat{b}_\lambda)$ minimize $\|\beta\|^2 + \lambda\|b\|^2$ subject to $U\beta + Vb = \mathbf{Y}$. Set $H = D^{-1}VV^\top$, $A = U^\top H^{-1}U$, and define the two-stage fit by

$$\tilde{\beta} = \arg \min_{\beta} (\mathbf{Y} - U\beta)^\top H^{-1}(\mathbf{Y} - U\beta), \quad \tilde{b} = V^\top (VV^\top)^{-1}(\mathbf{Y} - U\tilde{\beta}).$$

(a) Prove that

$$\tilde{\beta} = A^{-1}U^\top H^{-1}\mathbf{Y}, \quad \hat{\beta}_\lambda - \tilde{\beta} = -(I_m + (\lambda/D)A)^{-1}\tilde{\beta}.$$

If $A \succeq aI_m$ with $a > 0$, deduce

$$\|\hat{\beta}_\lambda - \tilde{\beta}\| \leq \frac{\|\tilde{\beta}\|}{1 + (\lambda/D)a}.$$

(b) Put $d_\lambda = \hat{\beta}_\lambda - \tilde{\beta}$. Prove that

$$\hat{b}_\lambda - \tilde{b} = -V^\top (VV^\top)^{-1}Ud_\lambda, \quad \|\hat{b}_\lambda - \tilde{b}\|^2 = D^{-1}d_\lambda^\top A d_\lambda.$$

For the lecture's choice $\lambda = D$, show that on $\mathcal{G}$,

$$\|\hat{\beta}_D - \tilde{\beta}\| \leq \frac{\|\tilde{\beta}\|}{1 + n/(2C_H)}, \quad \|\hat{b}_D - \tilde{b}\|^2 \leq \frac{2C_H}{Dn} \|\tilde{\beta}\|^2.$$

Distinguish the limit $\lambda \rightarrow \infty$ with $U, V, \mathbf{Y}$ fixed from the lecture's bound, where D, V , and n may vary together.

(c) Show that $\tilde{\beta}$ is the ordinary least-squares solution when H is a scalar multiple of I_n . For

$$U = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad V = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 2\sqrt{2} \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad D = 2,$$

calculate $\tilde{\beta}$ and the ordinary least-squares solution. Explain why a large second-block penalty alone does not imply convergence to ordinary least squares.

HINTS Eliminate b as in [Exercise 3](#). For the last bound in (b), diagonalize A and use $t/(1+t)^2 \leq 1/t$ for $t > 0$.

Exercise 8: Two derivations of the optimizer identities.

Derive the optimizer from its optimality equations and verify it with a push-through identity.

Let $U \in \mathbb{R}^{n \times m}$ and $V \in \mathbb{R}^{n \times D}$, where V has full row rank, and define

$$H = D^{-1}VV^\top, \quad K = H + UU^\top, \quad \alpha = K^{-1}\mathbf{Y}.$$

Assume that H is positive definite. Consider

$$\min_{\beta \in \mathbb{R}^m, b \in \mathbb{R}^D} \left\{ \frac{1}{2} \|\beta\|^2 + \frac{D}{2} \|b\|^2 : U\beta + Vb = \mathbf{Y} \right\}.$$

(a) Introduce a Lagrange multiplier in $\mathbb{R}^n$ and use the optimality equations to prove that the optimizer satisfies

$$\hat{\beta} = U^\top \alpha, \quad \hat{b} = D^{-1}V^\top \alpha.$$

(b) Let $A \in \mathbb{R}^{p \times q}$ and $B \in \mathbb{R}^{q \times p}$, and suppose the two inverses below exist. Starting from

$$(I_p + AB)A = A(I_q + BA),$$

prove the push-through identity

$$(I_p + AB)^{-1}A = A(I_q + BA)^{-1}.$$

Then apply it to prove

$$(I_m + U^\top H^{-1}U)^{-1}U^\top H^{-1} = U^\top (H + UU^\top)^{-1}.$$

Conclude that the formula for $\hat{\beta}$ in Eq. (9.18) equals $U^\top \alpha$.

(c) Starting from the formula for b_β in Eq. (9.17), verify directly that

$$\hat{b} = D^{-1}V^\top \alpha.$$

Your calculation should use $VV^\top = DH$ and $I_n - UU^\top K^{-1} = HK^{-1}$.

Remark 9.8.1 (Why “push-through”?). In $(I_p + AB)^{-1}A = A(I_q + BA)^{-1}$, the factor A moves through the inverse. The matrix being inverted changes from a $p \times p$ matrix to a $q \times q$ matrix. This identity is especially useful when one of these two matrices is substantially smaller than the other.

Exercise 9: Matrix inequalities in the interpolation proof.

☉ Justify the spectral and matrix-order bounds used in the risk calculation.

For symmetric matrices, $S \preceq R$ means $x^\top Sx \leq x^\top Rx$ for every vector x . The matrix norm is the Euclidean operator norm $\|M\| = \sup_{\|x\|=1} \|Mx\|$. You may use the spectral theorem: every real symmetric matrix has an orthonormal basis of eigenvectors.

(a) Prove that $S \preceq R$ implies $B^\top SB \preceq B^\top RB$ for any matrix B of compatible size. If $c_H I_n \preceq H \preceq C_H I_n$ with $c_H > 0$, prove

$$C_H^{-1} I_n \preceq H^{-1} \preceq c_H^{-1} I_n, \quad H^{-2} \preceq c_H^{-1} H^{-1}.$$

Identify which inequalities follow by applying scalar inequalities to the eigenvalues of the same symmetric matrix.

(b) Suppose also $U^\top U \succeq (n/2)I_m$ for $U \in \mathbb{R}^{n \times m}$. With $A = U^\top H^{-1}U$ and $T = (I_m + A)^{-1}U^\top H^{-1}$, prove

$$A \succeq \frac{n}{2C_H} I_m, \quad \|(I_m + A)^{-1}\| \leq \frac{2C_H}{n},$$

and

$$TT^\top \preceq c_H^{-1} (I_m + A)^{-1} A (I_m + A)^{-1} \preceq \frac{2C_H}{c_H n} I_m.$$

Explain why it is legitimate to diagonalize A and $(I_m + A)^{-1}$ simultaneously.

(c) Set $K = H + UU^\top$, $\alpha = K^{-1}\mathbf{Y}$, $H = D^{-1}VV^\top$, and $\hat{b} = D^{-1}V^\top \alpha$. Prove the full chain

$$\|\hat{b}\|^2 = D^{-1} \alpha^\top H \alpha \leq D^{-1} \mathbf{Y}^\top K^{-1} \mathbf{Y} \leq (c_H D)^{-1} \|\mathbf{Y}\|^2.$$

HINTS For (b), first write $TT^\top$ explicitly. For an eigenvalue $t > 0$ of A , use $t/(1+t)^2 \leq 1/t$.

Exercise 10: Operator norms, traces, and noise depending on the inputs.

☉ Turn matrix bounds into expectation bounds without assuming independent residuals.

Use the Euclidean operator norm. The trace $\text{tr}(M)$ of a square matrix is the sum of its diagonal entries.

(a) For $T \in \mathbb{R}^{m \times n}$, prove

$$\|T\|^2 = \|T^\top T\| = \|TT^\top\|, \quad \text{tr}(T^\top T) = \text{tr}(TT^\top).$$

If $TT^\top \preceq bI_m$ with $b \geq 0$, deduce that $\|Tz\|^2 \leq b\|z\|^2$ and $\text{tr}(TT^\top) \leq mb$.

(b) Let $(X_i, \varepsilon_i)_{i=1}^n$ be i.i.d. with $\mathbb{E}[\varepsilon_i | X_i] = 0$ and $\mathbb{E}[\varepsilon_i^2 | X_i] \leq \sigma^2$. Let $T \in \mathbb{R}^{m \times n}$ depend only on $X_{1:n}$, and let $\mathcal{G}$ be an input event. Set $T = 0$ on $\mathcal{G}^c$ and assume $TT^\top \preceq bI_m$ on $\mathcal{G}$ for a deterministic $b < \infty$. Writing $\varepsilon = (\varepsilon_i)_{i=1}^n$, prove

$$\mathbb{E}[\|T\varepsilon\|^2 | X_{1:n}] = \text{tr}(T^\top T \mathbb{E}[\varepsilon\varepsilon^\top | X_{1:n}]) \leq \sigma^2 \text{tr}(TT^\top).$$

Deduce $\mathbb{E}[\mathbb{I}\{\mathcal{G}\} \|T\varepsilon\|^2] \leq \sigma^2 mb$. State exactly where the i.i.d. assumption, the zero conditional mean, and the fact that $\mathcal{G}$ is an input event enter the calculation.

(c) Let $r \in L^2(\mu)$ be fixed, where μ is the law of X_i , and set $r_X = (r(X_i))_{i=1}^n$. Show that

$$\mathbb{E}[\|r_X\|^2] = n\|r\|_{L^2(\mu)}^2, \quad \mathbb{E}[\mathbb{I}\{\mathcal{G}\} \|Tr_X\|^2] \leq bn\|r\|_{L^2(\mu)}^2.$$

Explain why independence of T and r_X is unnecessary.

HINTS For (a), use the singular value decomposition or the nonzero eigenvalues of $T^\top T$ and $TT^\top$. For (b), expand the quadratic form as a finite sum and condition on all the inputs before averaging.

Exercise 11: Consistency with finite average noise variance.

☉ Extend the interpolation guarantee to unbounded conditional variances by controlling their tails.

Retain the uniform-input, i.i.d. regression setup and $\mathbb{E}[\varepsilon_i | X_i] = 0$, but replace the uniform conditional variance bound by

$$s^2(x) = \mathbb{E}[\varepsilon_i^2 | X_i = x], \quad \bar{\sigma}^2 = \mathbb{E}[s^2(X_i)] = \mathbb{E}[\varepsilon_i^2] < \infty.$$

For $z \in \mathbb{R}$, write $z_+ = \max\{z, 0\}$.

(a) Let $T \in \mathbb{R}^{m \times n}$ depend only on the inputs, set $T = 0$ on $\mathcal{G}^c$, and suppose $TT^\top \preceq (C_T/n)I_m$ on $\mathcal{G}$ for a fixed $C_T > 0$. Prove that, for every deterministic $a \geq 0$,

$$\mathbb{E}[\mathbb{I}\{\mathcal{G}\} \|T\varepsilon\|^2] \leq C_T \left(\frac{am}{n} + \mathbb{E}[(s^2(X_1) - a)_+] \right).$$

No independence between T and $s^2(X_i)$ may be assumed.

(b) Define the deterministic quantity

$$\mathcal{V}(t) = \inf_{a \geq 0} \{at + \mathbb{E}[(s^2(X_1) - a)_+]\}, \quad t \geq 0.$$

Use (a) and the proof of [Theorem 9.3.1](#) to show that, for $m + D \geq n$, $D \geq 16m$, and $\delta \in (0, 1)$, with probability at least $1 - Cn^2/D - m \exp(-cn/m^2) - \delta$,

$$\|\hat{f} - f^\star\|_{L^2(\mu)}^2 \leq \frac{C}{\delta} \left(A_m(f^\star) + \mathcal{V}(m/n) + \frac{\|f^\star\|_{L^2(\mu)}^2}{n^2} + \frac{n}{D} (\|f^\star\|_{L^2(\mu)}^2 + \bar{\sigma}^2) \right).$$

The constants $c, C > 0$ are universal. Show that if $s^2(X_1) \leq \sigma^2$ almost surely, then $\mathcal{V}(m/n) \leq \sigma^2 m/n$.

- (c) Prove that $\mathcal{V}(t) \rightarrow 0$ as $t \downarrow 0$. Deduce consistency in probability for $m_n = \max\{1, \lfloor n^{1/4} \rfloor\}$ and $D_n = n^3$ for every fixed $f^* \in L^2(\mu)$ with $\bar{\sigma}^2 < \infty$. Explain why finite average variance alone does not give a decay rate for the tail expectation in (a).

HINTS For (a), condition on the inputs and split each variance using $s^2 \leq a + (s^2 - a)_+$. Use the trace bound for the first term and $\|Te_i\|^2 \leq C_T/n$ for each term in the remainder. For (c), fix a while taking $t \downarrow 0$, and then take $a \rightarrow \infty$.

Exercise 12: Equal penalties can eliminate a nonzero signal.

☉ Use a Gram-matrix bound to diagnose failure of a minimum-norm interpolator.

Use the notation of [Section 9.3](#). Suppose that on events $\mathcal{G}_n$ with $\mathbb{P}(\mathcal{G}_n) \rightarrow 1$, the tail Gram matrix satisfies $H = D_n^{-1}VV^\top \succeq cI_n$ for a constant $c > 0$. Let $\hat{f}_n^{\text{eq}}$ be the minimum- $L^2(\mu)$ -norm polynomial that interpolates $\mathbf{Y}$. Assume $\mathbb{E}[Y^2] < \infty$ and $D_n/n \rightarrow \infty$.

- (a) On $\mathcal{G}_n$, construct an interpolating polynomial using only the Chebyshev features in v . Use [Eq. \(9.9\)](#) and the minimality of $\hat{f}_n^{\text{eq}}$ to prove

$$\|\hat{f}_n^{\text{eq}}\|_{L^2(\mu)}^2 \leq \frac{\pi}{2cD_n} \|\mathbf{Y}\|^2.$$

- (b) Prove $\|\hat{f}_n^{\text{eq}}\|_{L^2(\mu)} \xrightarrow{\mathbb{P}} 0$ as $n \rightarrow \infty$. Then show that for every fixed nonzero $f^* \in L^2(\mu)$,

$$\|\hat{f}_n^{\text{eq}} - f^*\|_{L^2(\mu)}^2 \xrightarrow{\mathbb{P}} \|f^*\|_{L^2(\mu)}^2 \quad \text{as } n \rightarrow \infty.$$

- (c) The estimator interpolates every sample with distinct inputs. Explain why this fact is compatible with convergence to zero in population norm. Your answer must distinguish values on the random training set from an $L^2(\mu)$ statement about test inputs.

HINTS For (a), solve $Vb = \mathbf{Y}$ with its minimum-Euclidean-norm coefficient vector. For (b), apply the weak law of large numbers to $n^{-1}\|\mathbf{Y}\|^2$ and Cauchy-Schwarz to the cross term in the squared distance.

9.A Background: L^2 , orthogonality, and projections

The population norm measures functions by their average squared values under the input law. The resulting geometry lets us separate approximation error from coefficient error.

For a probability measure μ , the space $L^2(\mu)$ consists of real measurable functions f with $\int f^2 d\mu < \infty$, identifying two functions when they agree μ -almost everywhere. Define

$$\langle f, g \rangle_{L^2(\mu)} = \int fg d\mu, \quad \|f\|_{L^2(\mu)} = \left(\int f^2 d\mu \right)^{1/2}.$$

An *inner product* is a symmetric bilinear form that is positive on every nonzero vector; it induces a norm by $\|f\| = \sqrt{\langle f, f \rangle}$. Identifying functions that agree almost everywhere is necessary here: the integral of f^2 can be zero even if f is nonzero on a set of measure zero. Thus $L^2(\mu)$ does not distinguish changes at finitely many points when μ has no atoms. Interpolation constraints concern the values of chosen function representatives, while population error concerns their $L^2(\mu)$ classes.

A *Hilbert space* is an inner-product space complete in its induced norm: every Cauchy sequence converges to an element of the space. $L^2(\mu)$ is a Hilbert space. Its basic inequalities are

$$|\langle f, g \rangle_{L^2(\mu)}| \leq \|f\|_{L^2(\mu)} \|g\|_{L^2(\mu)}, \quad \|f + g\|_{L^2(\mu)} \leq \|f\|_{L^2(\mu)} + \|g\|_{L^2(\mu)}.$$

The first is Cauchy–Schwarz. The second follows by expanding $\|f + g\|_{L^2(\mu)}^2$ and applying the first.

Two functions are *orthogonal* when their inner product is zero. For such functions, expansion of the squared norm gives Pythagoras:

$$\|f + g\|_{L^2(\mu)}^2 = \|f\|_{L^2(\mu)}^2 + \|g\|_{L^2(\mu)}^2.$$

A family (ψ_j) is *orthonormal* when $\langle \psi_j, \psi_k \rangle_{L^2(\mu)} = \mathbb{I}\{j = k\}$. It is an *orthonormal basis* if its finite linear combinations are dense: every element of $L^2(\mu)$ can be approximated arbitrarily well in norm by those combinations. Such a basis can be infinite, and a general function is the norm limit of its expansions. For the finite span of $\psi_0, \dots, \psi_{m-1}$, the *orthogonal projection* of f is

$$g_m = \sum_{j < m} \langle f, \psi_j \rangle_{L^2(\mu)} \psi_j.$$

It is the unique best approximation in that span, and $f - g_m$ is orthogonal to every element of the span. In [Exercise 5](#), you are asked to prove these claims and the resulting norm identities. A finite-dimensional subspace is closed, so the best approximation exists; completeness of the ambient Hilbert space also guarantees projection onto arbitrary closed subspaces.

9.B Why the polynomial matrices satisfy the required bounds

Why do the actual polynomial features satisfy the matrix conditions used in the main proof? Their coordinates are dependent, so an independent-coordinate argument would not answer this question. We use their orthogonality, boundedness, and the independence of the sampled inputs.

Lemma 9.B.1 (Sample matrices for the two polynomial bases). *For the uniform input law and features in [Section 9.3](#), if $D \geq 16m$, there are universal constants $c_H, C_H, c_{\text{pr}}, C_{\text{pr}} > 0$ and an input event $\mathcal{G}$ such that*

$$c_H I_n \preceq H \preceq C_H I_n, \quad U^\top U \succeq (n/2) I_m$$

on $\mathcal{G}$, and

$$\mathbb{P}(\mathcal{G}^c) \leq C_{\text{pr}} n^2 / D + m \exp(-c_{\text{pr}} n / m^2).$$

Moreover, $\Sigma_v := \mathbb{E}[v(X)v(X)^\top] \preceq (\pi/2) I_D$.

Proof. We verify the three claims in turn.

The interpolation Gram matrix. How can many dependent polynomial features yield a well-conditioned $n \times n$ matrix? Its diagonal stays bounded, while its off-diagonal entries become small in mean square.

Put $Q_q^x(x) = \sum_{j < q} \chi_j(x)^2$. For every $q \geq 1$ and $x \in [-1, 1]$,

$$q/4 \leq Q_q^x(x) \leq 2q. \tag{9.33}$$

The upper bound uses $|\chi_j| \leq \sqrt{2}$. To prove the lower bound, write $x = \cos \theta$. When $\sin \theta \geq 2/q$,

$$Q_q^x(\cos \theta) = q - \frac{1}{2} + \frac{\sin((2q-1)\theta)}{2 \sin \theta} \geq q - \frac{1}{2} - \frac{q}{4} \geq \frac{q}{4}.$$

This case can occur only for $q \geq 2$. Otherwise let $\vartheta = \min(\theta, \pi - \theta)$. The inequality $\sin \vartheta \geq 2\vartheta/\pi$ implies $\vartheta < \pi/q$. For $1 \leq j \leq \lfloor q/4 \rfloor$, $2\cos^2(j\vartheta) \geq 1$. Including the constant feature gives $Q_q^\chi \geq 1 + \lfloor q/4 \rfloor \geq q/4$.

The diagonal of H is $d(X_i)$, where

$$d(x) = D^{-1}(Q_{m+D}^\chi(x) - Q_m^\chi(x)).$$

By Eq. (9.33), if $D \geq 16m$, then

$$1/8 \leq d(x) \leq 2.$$

Let $J = \text{diag}(d(X_1), \dots, d(X_n))$ and $E = H - J$. The domination $\mu \leq (\pi/2)\nu$ and orthonormality under ν give $\Sigma_v \preceq (\pi/2)I_D$. For $i \neq \ell$, independence of X_i, X_ℓ therefore gives

$$\mathbb{E}[H_{i\ell}^2] = D^{-2} \text{tr}(\Sigma_v^2) \leq \frac{\pi^2}{4D}.$$

Consequently $\mathbb{E}[\|E\|_F^2] \leq Cn^2/D$. Markov's inequality and $\|E\| \leq \|E\|_F$ show that $\|E\| \leq 1/16$ except on an event of probability at most Cn^2/D . On this event,

$$(1/16)I_n \preceq H \preceq (33/16)I_n.$$

The low-degree design matrix. Why are its m coefficients estimable from n samples? Legendre orthonormality gives $\mathbb{E}[u(X)u(X)^\top] = I_m$, and $|P_j| \leq 1$ gives

$$\|u(x)\|^2 = \sum_{j < m} (2j+1)P_j(x)^2 \leq m^2.$$

The matrix Chernoff inequality for independent positive semidefinite matrices $Z_i \in \mathbb{R}^{m \times m}$ with $0 \preceq Z_i \preceq RI_m$ states that

$$\mathbb{P}\left(\lambda_{\min}\left(\sum_i Z_i\right) \leq \frac{1}{2}\lambda_{\min}\left(\sum_i \mathbb{E}[Z_i]\right)\right) \leq m \exp\left(-c \lambda_{\min}\left(\sum_i \mathbb{E}[Z_i]\right)/R\right).$$

Apply it to $Z_i = u(X_i)u(X_i)^\top$ with $R = m^2$ and $\sum_i \mathbb{E}[Z_i] = nI_m$. This proves $U^\top U \succeq (n/2)I_m$ outside an event of probability $m \exp(-cn/m^2)$. Intersecting the two input events proves the lemma. $\square$

The main proof uses only this lemma, orthonormality of the low-degree features, and the synthesis bound for the additional features. Thus the argument applies to other series whenever those same properties hold. The polynomial calculation provides a concrete verification of those properties on a fixed, one-dimensional problem.

9.C Expected-risk failure and its remedies

Does the positive probability guarantee also control the average loss over training samples? For our unmodified polynomial interpolator the answer is no. The obstruction is not specific to polynomials: fitting different noisy responses at nearby inputs forces large variation, and some function spaces can realize that variation only at large population norm. We first state that general principle, then show what clipping repairs.

9.C.1 A general obstruction from nearby inputs

Which property of the function space makes close observations dangerous? Let $(\mathcal{X}, d)$ be a metric space and let X_1, X_2 be independent with law μ . For the next result let $\mathcal{F}_n \subset L^2(\mu)$.

Proposition 9.C.1 (Interpolation with norm-controlled variation). *Suppose $n \geq 2$, $\mathbb{E}[Y^2] < \infty$, and the i.i.d. observations satisfy*

$$\mathbb{E}[Y_i | X_i] = f^*(X_i), \quad \text{Var}(Y_i | X_i) \geq \sigma_0^2 > 0.$$

Assume that for some deterministic $0 < C_n < \infty$ and measurable function $\omega : (0, \infty) \rightarrow (0, \infty)$,

$$|g(x) - g(x')| \leq C_n \omega(d(x, x')) \|g\|_{L^2(\mu)} \quad (g \in \mathcal{F}_n, x \neq x'), \quad (9.34)$$

and that

$$\mathbb{P}(X_1 = X_2) = 0, \quad \mathbb{E}[\omega(d(X_1, X_2))^{-2}] = \infty.$$

Every measurable estimator $\hat{f}_n \in \mathcal{F}_n$ that interpolates the sample almost surely satisfies

$$\mathbb{E}[\|\hat{f}_n - f^*\|_{L^2(\mu)}^2] = \infty.$$

Proof. On the probability-one event of interpolation and $X_1 \neq X_2$, apply [Eq. \(9.34\)](#) to $g = \hat{f}_n$:

$$\begin{aligned} (Y_1 - Y_2)^2 &= |\hat{f}_n(X_1) - \hat{f}_n(X_2)|^2 \\ &\leq C_n^2 \omega(d(X_1, X_2))^2 \|\hat{f}_n\|_2^2. \end{aligned}$$

The factor $C_n^2 \omega(d(X_1, X_2))^2$ is strictly positive, so division gives

$$\|\hat{f}_n\|_2^2 \geq \frac{(Y_1 - Y_2)^2}{C_n^2 \omega(d(X_1, X_2))^2}.$$

Independence of the two observations gives, conditional on their inputs,

$$\mathbb{E}[(Y_1 - Y_2)^2 | X_1, X_2] \geq (f^*(X_1) - f^*(X_2))^2 + 2\sigma_0^2 \geq 2\sigma_0^2.$$

Taking expectations of the nonnegative quantities gives

$$\mathbb{E}[\|\hat{f}_n\|_2^2] \geq \frac{2\sigma_0^2}{C_n^2} \mathbb{E}[\omega(d(X_1, X_2))^{-2}] = \infty.$$

Finally, $\|\hat{f}_n - f^*\|_2^2 \geq \frac{1}{2} \|\hat{f}_n\|_2^2 - \|f^*\|_2^2$. □

In the applications below, ω is nondecreasing and therefore Borel measurable. For an arbitrary measurable modulus, replacing ω by the nondecreasing majorant $\bar{\omega}(t) = \sup_{0 < u \leq t} \omega(u)$ is not automatic: when this majorant is finite, it preserves [Eq. \(9.34\)](#), but the inequality $\bar{\omega}^{-2} \leq \omega^{-2}$ can destroy the required infinite inverse-square expectation. The two conditions must therefore be checked for the chosen modulus.

To express the consequence in terms of the benign set, let $\mathcal{A}_E(B)$ denote the version of [Eq. \(9.2\)](#) requiring $\mathbb{E}[\|B_n(S_n) - f_P^*\|_{L^2(P_X)}^2] \rightarrow 0$ as $n \rightarrow \infty$ instead of convergence in probability. If a fixed law P and the interpolating procedure's output classes satisfy the proposition for all sufficiently large n , then $P \notin \mathcal{A}_E(B)$. The close-input condition is quantitative: the expected inverse-square modulus must diverge, not merely be large. This exclusion does not imply $P \notin \mathcal{A}(B)$ with its original definition.

For uniform inputs on $[-1, 1]$, take $\omega(t) = t$. Then $\mathbb{E}[|X_1 - X_2|^{-2}] = \infty$.

Proposition 9.C.2 (Variation control in finite-dimensional smooth spaces). *Let μ be the uniform law on $[-1, 1]$, and let $\mathcal{F}_n \subset C^1([-1, 1])$ be a fixed finite-dimensional linear subspace. There is a constant $0 < C_n < \infty$ such that*

$$|g(x) - g(x')| \leq C_n |x - x'| \|g\|_{L^2(\mu)} \quad (g \in \mathcal{F}_n, x, x' \in [-1, 1]).$$

Thus [Eq. \(9.34\)](#) holds with $\omega(t) = t$.

Proof. Continuity ensures that $\|g\|_{L^2(\mu)} = 0$ implies $g = 0$ everywhere, so $\|\cdot\|_{L^2(\mu)}$ is a norm on $\mathcal{F}_n$. By finite-dimensional norm equivalence, the norm $\|g\|_{C^1} := \|g\|_\infty + \|g'\|_\infty$ satisfies

$$\|g'\|_\infty \leq \|g\|_{C^1} \leq C_n \|g\|_{L^2(\mu)} \quad (g \in \mathcal{F}_n).$$

The mean-value theorem now gives $|g(x) - g(x')| \leq |x - x'| \|g'\|_\infty$, proving the claim. $\square$

This includes degree-at-most- p_n polynomials for every deterministic finite p_n , even when p_n grows very rapidly with n . It also includes finite spans of fixed continuously differentiable features. The constant C_n may grow with dimension; its finiteness at each n is enough.

In particular, the result applies to the exact procedure in [Theorem 9.3.1](#). It does not apply to every smooth interpolating method: a class permitting arbitrarily narrow corrections need not satisfy [Eq. \(9.34\)](#) with one deterministic finite C_n . For example, bounded piecewise-linear interpolation can accommodate a close pair by placing its steep segment only between the pair.

9.C.2 What infinite expectation signals

How can risk tend to zero in probability while its expectation is infinite? The exceptional samples have probabilities tending to zero, but their losses have no integrable uniform bound. The example below makes this failure visible without a complicated target or unbounded observation noise.

Take $f^* = 0$ and independent Rademacher noise $Y_i \in \{-1, 1\}$. For any fixed n and class satisfying $|g(x) - g(x')| \leq C_n |x - x'| \|g\|_2$, the event $Y_1 \neq Y_2$ has probability $1/2$. On it,

$$\|\hat{f}_n\|_2^2 \geq \frac{4}{C_n^2 |X_1 - X_2|^2}.$$

For independent uniform inputs, $\mathbb{P}(|X_1 - X_2| \leq t) = t - t^2/4$ for $0 \leq t \leq 2$. Hence, for all sufficiently large z ,

$$\mathbb{P}(\|\hat{f}_n\|_2^2 > z) \geq \frac{c}{C_n \sqrt{z}}.$$

Integrating this tail over z diverges. The large loss comes from the estimator, although the target and responses are bounded. In a finite-dimensional smooth space, fitting a steep change at a close pair forces a large L^2 norm somewhere on the domain. The mean-value theorem alone locates a large derivative, not the location of the large predictions.

This matters when repeated retraining is evaluated by average squared loss, or when occasional extreme predictions are unacceptable. The proposition is a tail-risk result. Finite measurement precision removes the literal arbitrarily-close-pair idealization, but can still leave severe sensitivity. A probability guarantee and a tail safeguard therefore answer different practical questions.

9.C.3 Clipping and regularization

Can we control the rare failures without losing the positive result? Suppose a bound $|f^*(x)| \leq M$ is known, and define the clipped predictor

$$\bar{f}_n(x) = [\hat{f}_n(x)]_{[-M, M]} := \max\{-M, \min\{M, \hat{f}_n(x)\}\}.$$

Projection onto an interval cannot increase the distance to a point of that interval. Thus, sample by sample,

$$\|\bar{f}_n - f^*\|_2^2 \leq \|\hat{f}_n - f^*\|_2^2, \quad \|\bar{f}_n - f^*\|_2^2 \leq 4M^2.$$

This already turns consistency in probability into expected-risk consistency. For our procedure, the proof gives a rate as well.

Proposition 9.C.3 (Clipping recovers the expected-risk rate). *Under the assumptions of Theorem 9.3.1, suppose $|f^*| \leq M$ μ -almost everywhere. Then*

$$\begin{aligned} \mathbb{E}[\|\bar{f}_n - f^*\|_2^2] &\leq C \left(A_m(f^*) + \frac{\sigma^2 m}{n} + \frac{\|f^*\|_2^2}{n^2} + \frac{n}{D} (\|f^*\|_2^2 + \sigma^2) \right) \\ &\quad + 4M^2 (Cn^2/D + m \exp(-cn/m^2)). \end{aligned} \quad (9.35)$$

For the choices in Corollary 9.3.3, this is $O(n^{-2s/(2s+1)})$. If $|Y| \leq M$ almost surely as well, $\bar{f}_n$ still interpolates.

Proof. Split the expectation on $\mathcal{G}$ and $\mathcal{G}^c$. On $\mathcal{G}$, clipping cannot increase the error, so use the expectation estimates Eqs. (9.16), (9.22) and (9.23) from the main proof. On $\mathcal{G}^c$, the error is at most $4M^2$. This proves Eq. (9.35). When $D_n = n^3$, the new contribution $n^2/D_n = n^{-1}$ is smaller than the claimed rate, and the design failure term decays faster than any power. Finally, if $|Y_i| \leq M$, clipping leaves $\hat{f}_n(X_i) = Y_i$ unchanged. $\square$

A known response range is stronger than a known range for the conditional mean. With bounded responses, clipping preserves interpolation but generally leaves the polynomial space, so the prediction rule is the minimum-norm fit followed by clipping. With only a target bound, some noisy responses may lie outside the interval, and clipping no longer interpolates those observations. Both are valid remedies, but they define different procedures and different benign sets.

Positive ridge regularization is another remedy: it permits residual training error instead of insisting on fitting an unstable sample exactly. For a fixed bounded feature map ϕ and

$$\hat{\theta}_\lambda \in \arg \min_{\theta} \left\{ \frac{1}{n} \sum_i (\theta^\top \phi(X_i) - Y_i)^2 + \lambda \|\theta\|^2 \right\}, \quad \lambda > 0,$$

comparison with $\theta = 0$ gives $\|\hat{\theta}_\lambda\|^2 \leq \|\mathbf{Y}\|^2/(n\lambda)$. If $\sup_x \|\phi(x)\|^2 \leq K_n < \infty$ and $\mathbb{E}[Y^2] < \infty$, then

$$\mathbb{E}[\|\hat{\theta}_\lambda^\top \phi\|_{L^2(\mu)}^2] \leq K_n \mathbb{E}[Y^2]/\lambda < \infty.$$

This removes the fixed-sample infinite expectation. Choosing λ_n to obtain consistency or an optimal rate is a further statistical question; positivity alone does not establish either.

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